
COMPARATIVE STUDY OF POTTS MACHINE DYNAMICS AND PERFORMANCE FOR MAX- k -CUT

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ABSTRACT

We present the first systematic comparison of analog Potts machines (PMs), which are multi-state extensions of Ising machines (IMs), applied to solving Max-3-Cut and Max-4-Cut problems. We analyze three existing PM models: nonequilibrium condensate (NEC), q -photon downconversion (q -PDC), and q^{th} sub-harmonic injection locking (q -SHIL, corresponding to the Kuramoto model). We also propose two new ones: polynomial and sigmoid PMs. We also include an analog IM model for reference. We find a clear link between performance and three dynamical properties: a connected origin-to-solution fixed-point branch, enforced amplitude homogeneity, and enforced phase discretization. Our benchmark on problems from the GSet collection indicate that in terms of optimality gap, the NEC model performs best among the PMs, followed by the sigmoid and the polynomial models. Yet, the reference IM exhibits lower optimality gaps than all PMs. We also observe a significant reduction in performance in all PMs when moving from Max-3-Cut to Max-4-Cut, indicating a mismatch between current PM formulations and the standard Potts model. Despite these findings, PMs remain promising for analog implementations because they directly encode multi-state variables without the overhead of binary expansions. Our results highlight the importance of dynamical properties in guiding high-performance PM designs and suggest new research directions for analog solvers for combinatorial optimization.

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1 Introduction

Combinatorial optimization is concerned with minimizing a cost function over a set of discrete variables. Such problems are key to a wide range of applications, including machine learning, chip design, robotics, manufacturing, and operations research. The size of the configuration space of combinatorial optimization problems (COPs) can grow faster than exponentially in the number of objects, making exhaustive search impractical [1]. As a result, over the years, several algorithms and hardware accelerators have been developed aiming to provide high-quality solutions with minimal consumption of resources such as time and energy.

Ising machines (IMs) [2, 3, 4, 5] are hardware accelerators designed to find low-energy states of the *Ising model* of statistical mechanics, defined by the Ising Hamiltonian [6]:

$$H_{\text{Ising}}(\{\sigma_i\}) = -\frac{1}{2} \sum_{i,j}^N J_{ij} \sigma_i \sigma_j - \sum_i^N h_i \sigma_i. \quad (1)$$

Here, $\sigma_i \in \{-1, +1\}$ represent binary spins such as the magnetic moment of a particle, which can either be spin-up (+1) or spin-down (-1). The first term describes the interaction between spins, where J_{ij} is the real-valued coupling coefficient between spins i and j , and the second term describes the real-valued external bias field h_i acting on each spin. The goal of an IM is to find a configuration of spins that minimizes the energy of the system, which corresponds to finding a ground state of the Ising Hamiltonian.

While IMs have been widely studied as analog computing platforms for solving many NP-hard COPs [7], many real-world and industry-relevant COPs are naturally defined using multi-level discrete variables. Max- k -Cut is an archetypal example of such a problem, where the objective is to partition the vertices of a graph into k sets such that the number of edges between different sets is maximized. This can be described as maximizing the following objective function, referred to as the *cut value*:

$$C(\{v_i\}) = \frac{1}{2} \sum_{i,j}^N w_{ij} (1 - \delta(v_i, v_j)). \quad (2)$$

Here, $v_i \in \{1, 2, \dots, k\}$ is the label denoting which set vertex i belongs to, w_{ij} is the weight of the edge between vertices i and j , and $\delta(s_i, s_j)$ is the Kronecker delta, which equals 1 if $s_i = s_j$ and 0 otherwise.

For $k = 2$, Max- k -Cut reduces to the Max-Cut problem, which maps directly to the Ising model by setting $\sigma_i = 2s_i - 3$, $J_{ij} = -w_{ij}$, and $h_i = 0$, and can readily be addressed by IMs. However, for $k > 2$, Max- k -Cut cannot be natively represented by binary variables, and requires a binary expansion in order to map the problem to the Ising model [7]. Such expansion incurs significant overhead in terms of embedding size, in turn increasing hardware complexity and power consumption. Furthermore, the embedding overhead can lead to a less convex

energy landscape, resulting in sub-optimal performance. A similar issue has been identified k -SAT problems, where polynomial unconstrained binary optimization (PUBO) outperforms quadratization methods [8, 9, 10, 11] due to the latter providing a less native mapping of the original problem structure.

To address the limitations in mapping multi-level discrete variable COPs to the Ising model, Potts machines (PMs) have been proposed as extensions of IMs that directly encode multi-state discrete variables [12, 13, 14, 15, 16]. PMs are designed to find low-energy states of the Potts model, which comes in two variants: the *standard Potts model* and the *planar Potts model*, defined by the standard Potts Hamiltonian and the planar Potts Hamiltonian, respectively [17]:

$$H_{\text{Standard Potts}}(\{s_i\}) = -\frac{1}{2} \sum_{i,j}^N J_{ij} \delta(s_i, s_j), \quad (3a)$$

$$H_{\text{Planar Potts}}(\{\theta_i\}) = -\frac{1}{2} \sum_{i,j}^N J_{ij} \cos(\theta_i - \theta_j). \quad (3b)$$

For the standard Potts model, $s_i \in \{1, 2, \dots, q\}$ is the state of spin i , while for the planar Potts model, θ_i is the state of spin i , taking one of q equispaced angles in the range $[-\pi, \pi)$. The two Potts models include a coupling term with real-valued coupling coefficients J_{ij} . The two Hamiltonians are equivalent for $q \leq 3$, where the angles can be mapped to the discrete states of the standard Potts model. For $q = 2$, both variants reduce to the Ising Hamiltonian defined in Eq. (1), without an external bias field, i.e. $h_i = 0$. The goal of a PM is to find a configuration of spins that minimizes the energy of the Potts model, which corresponds to finding a ground state of a Potts Hamiltonian.

Max- k -Cut can be mapped directly to the standard Potts model, simply by setting $q = k$, $s_i = v_i$ and $J_{ij} = -w_{ij}$, such that maximizing the cut value and minimizing the Potts Hamiltonian are equivalent. Max- k -Cut is thus the canonical problem for PMs, allowing PMs to solve these problems without embedding overhead. However, for $q > 3$, the planar and standard Potts Hamiltonians are not equivalent, so PMs implementing the planar Potts model will introduce additional constraints on the problem structure which can limit performance, as will be discussed in the Discussion section 6.4.

While the field of Ising machines has seen extensive studies of how the computational performance compares across different implementations [18, 19] and studies of the properties that are conducive to high performance [20], no such studies have been conducted for Potts machines as of yet. Many PMs have been proposed thus far, which we group into two categories: (i) discrete-spin PMs, where the spins directly take the discrete states of the Potts model, such as in implementations using probabilistic multi-dimensional bits (p-dits) [21] and single photon avalanche diodes (SPADs) [22], or (ii) continuous-spin

PMs, also called *analog PMs*, where the spins take continuous values and typically evolve dynamically to approach the discrete states of the Potts model, such as in implementations using physical coupled multi-stable gain-dissipative systems [12, 13, 14, 15, 16].

This work constitutes, to the best of our knowledge, the first comparative study of PMs, both qualitatively and quantitatively. We study only analog continuous-time PMs, which can all be described by coupled ordinary differential equations and simulated within a unified dynamical systems framework. We consider three such PM implementations from the literature: The nonequilibrium condensate (NEC) model by Kalinin and Berloff [12, 23], the q -photon downconversion (q -PDC) model by Honari-Latifpour et al. [24, 13], and the q^{th} sub-harmonic injection locking (q -SHIL) model by Roychowdhury and Seal [14]. Our study currently does not include the q -level phase sensitive amplifier (q -PSA) model by Inoue et al. [15], as it is a discrete-time model that does not fit our continuous-time framework. We also do not include the oscillator-inspired models by Mallick et al. [16] and Crnkic et al. [25] in our study, as they do not describe physical systems, but rather mathematical models not tied to a hardware implementation. Aside from the three PM models from the literature, we also propose two new PM models: a polynomial PM that unifies common traits among literature models, and a sigmoid PM drawing on insights from analog IMs [18, 7]. We also include an analog IM model [26] to study the performance of PMs relative to IMs.

Through theoretical analysis and numerical simulations, we study the dynamics of the five PM models and the IM model and identify three analytical requirements describing the dynamical behavior of the models: (i) a connected origin-to-solution fixed-point branch, (ii) enforced amplitude homogeneity, and (iii) complete phase discretization. Subsequently, we perform a quantitative study of the models over a benchmark set of Max-3-Cut and Max-4-Cut problems from the GSet collection to compare the performance of the PMs and IM in absolute terms.

By comparing benchmark performance to the dynamical analysis, we assess which dynamical properties are key to the performance of Potts machines for solving Max- k -Cut. These requirements can be used to guide the development of high-performance Potts machines, for instance as normative standards to determine which approaches hold promise.

2 Review of studied models, their governing equations and annealing schemes

The studied models are summarized in Table 1 and their governing equations are described in the following.

2.1 Non-Equilibrium Condensate (NEC) model

The NEC model [12, 23] is a continuous-time PM based on the dynamics of a Bose-Einstein condensate brought out of

equilibrium through a combination of resonant and nonresonant pumping. Kalinin and Berloff describe these dynamics by the complex Ginzburg-Landau equation (CGLE) in the presence of resonant pumping and dynamical gain. Assuming the single-spin coupling strengths are initialized as the coupling coefficients of the Potts Hamiltonian and assuming no in-site particle interactions, the governing equations of the NEC model can be rescaled as (see Supplementary Materials section S1.1 for details):

$$\frac{dx_i}{dt} = \alpha_i x_i - x_i |x_i|^{n-1} + \gamma (x_i^*)^{q-1} + \sum_{j=1}^N J_{ij} x_j, \quad (4a)$$

$$\frac{d\alpha_i}{dt} = \varepsilon_\alpha (\sqrt{r_{\text{target}}} - \sqrt{|x_i|}). \quad (4b)$$

The first equation describes the evolution of the complex-valued amplitude $x_i \in \mathbb{C}$ of condensate i , where α_i is the local gain, γ is the resonant pumping strength at the q^{th} resonance (with q corresponding to the number of Potts states, i.e. $q = 3$ for Max-3-Cut and $q = 4$ for Max-4-Cut), n is the nonlinearity order, N is the number of condensates, and J_{ij} are the coupling coefficients. The second equation describes the evolution of the local gain α_i , which is controlled by the gain relaxation rate ε_α and a target amplitude r_{target} . The nonlinearity order n is set to 3 in the original model. However, $n \geq q$ is required to ensure bounded amplitudes (see Supplementary Materials section S4). Therefore we employ $n = 3$ for Max-3-Cut ($q = 3$) and $n = 4$ for Max-4-Cut ($q = 4$).

The annealing scheme of the NEC model entails increasing the resonant pumping strength γ . Meanwhile, the local gains α_i evolve over time as described by Eq. (4b). The local gains are all initialized to $\alpha_i = -\mu_{\text{max}}$, where μ_{max} is the principal eigenvalue of the coupling matrix $\mathbf{J}$, which is found to be the threshold gain above which the origin becomes unstable (see Supplementary Materials section S2.1.1 for the stability analysis). The speed parameter ε_α is set to a small value (10^{-2}) for adiabatic operation with respect to Eq. (4a). We increase the resonant pumping strength γ linearly from zero over the course of the annealing with a rate $\varepsilon_\gamma = d\gamma/dt$, which is set to a small value (around 10^{-3} , see Supplementary Materials section S7) to ensure adiabatic annealing.

2.2 q -photon downconversion (q -PDC) model

The q -PDC model [24, 13] describes a Potts machine based on a network of phase-multistable coupled parametric oscillators. Phase multistability is achieved through the nonlinear process of spontaneous multiphoton downconversion, which is a process where a single pump photon is converted into q photons of lower energy. Assuming all spins have roughly equal number of couplings and equal coupling coefficients, the governing equations of the q -PDC model can be rescaled as (see Supplementary Materials section

Table 1: Summary of studied Potts machine (PM) and Ising machine (IM) models.

Model	Governing equations	Spin value	Amplitude nonlinearity	Annealing scheme	References
NEC	(4a), (4b)	$x \in \mathbb{C}$	Polynomial	Dynamical α , adiabatic γ increase	[23, 12]
q -PDC	(5a)	$x \in \mathbb{C}$	Polynomial	Fixed β , non-adiabatic (fast) γ increase	[24, 13]
q -SHIL	(6)	$\theta \in \mathbb{R}$	N/A	Adiabatic γ increase	[14]
Polynomial PM	(7)	$x \in \mathbb{C}$	Polynomial	Adiabatic β and γ increase	This work
Sigmoid PM	(8a), (8b)	$x \in \mathbb{C}$	Sigmoid	Fixed α , adiabatic β and γ increase	This work
Sigmoid IM	(9a)	$x \in \mathbb{R}$	Sigmoid	Fixed α , adiabatic β increase	[26, 7, 18]

The governing equations are provided in the text, while further details on the models and their governing equations, including parameter rescaling and stability analyses, are provided in the Supplementary Materials. The polynomial PM and sigmoid PM are new models proposed in this work. The sigmoid IM employs the spin sign method [26] and is included for comparison with the Potts machines.

S1.2 for details):

$$\frac{dx_i}{dt} = -x_i - x_i |x_i|^{2(q-1)} - \gamma (x_i^*)^{q-1} + \beta \sum_j^N J_{ij} x_j, \quad (5a)$$

$$\beta = \frac{1}{\kappa_d}, \quad \kappa_d = \kappa_l + \frac{1}{N} \sum_{ij} |J_{ij}|, \quad (5b)$$

where $x_i \in \mathbb{C}$ is the complex-valued amplitude of spin i , γ is the small-signal gain, β is the inverse of the oscillator damping rate κ_d , q is the number of photons produced in the downconversion process, and J_{ij} are again the coupling coefficients. The damping rate κ_d is the sum of the intrinsic loss of the oscillators κ_l and the average external losses due to interactions with other spins. We will use the value $\kappa_l = 1$ for all simulations, which is the value used in the original model [24].

The q -PDC model is annealed by linearly increasing γ , starting from zero, with a rate $\varepsilon_\gamma = d\gamma/dt$. The inverse damping rate β is fully determined by κ_l and the coupling matrix and is thus constant throughout annealing. If $\beta < \mu_{\max}^{-1}$, where $\mu_{\max}$ is the principal eigenvalue of the coupling matrix, the origin ($x_i = 0 \forall i$) remains a stable equilibrium (attracting) throughout annealing (see Supplementary Materials section S2.1.1). This means that the system can get trapped in the origin indefinitely. Instead of relying on destabilization of the origin, the oscillators are seeded at a non-zero amplitude and γ is increased fast enough to allow the system to reach a stable fixed point occurring through a bifurcation at threshold gain $\gamma_{\text{th}} = \frac{4}{3^{3/4}} \approx 1.755$ [24] before getting trapped in the origin. As γ must be annealed fast enough to avoid trapping, this model does not support adiabatic annealing, except for the case of $\beta > \mu_{\max}^{-1}$, where the origin is always unstable (repelling). We will employ a seeding amplitude of $r_{\text{seed}} = 1$, which is the value used in the original model [24]. For the problems studied in this work, an annealing rate of around $0.5\gamma_{\text{th}} \approx 1$ is sufficient to ensure that the system reaches a stable fixed point before getting trapped in the origin.

2.3 q -th sub-harmonic injection locking (q -SHIL) model

The q -SHIL model [14] describes a PM implemented as a system of coupled phase oscillators perturbed by an oscillation at the q^{th} harmonic of the oscillator frequency. The governing equations of the q -SHIL model can be rescaled as (see Supplementary Materials section S1.3 for details), corresponding the Kuramoto model [27] with a q -th harmonic perturbation:

$$\frac{d\theta_i}{dt} = -\gamma \sin q\theta_i + \sum_{j=1}^N J_{ij} \sin(\theta_j - \theta_i), \quad (6)$$

where $\theta_i \in \mathbb{R}$ is the phase of spin i , and γ is the injection locking strength. Note that this model corresponds to a simplification of the q -PDC model proposed by Honari-Latifpour et al. [13] assuming fixed, homogeneous amplitudes.

The q -SHIL model is annealed by linearly increasing γ , starting from zero, with a rate $\varepsilon_\gamma = d\gamma/dt$, which we set to 2×10^{-3} for adiabatic annealing. Note that there is no notion of an origin in this model, as it just describes phase dynamics for unit-amplitude spins.

2.4 Polynomial Potts machine model (Polynomial PM) - this work

The polynomial PM model is a new model proposed in this work, which both serves as a unifying model for NEC and q -PDC, assuming prefactors of the governing equations are homogeneous across all spins (so no local gain as in the NEC model), and as a Potts extension of the coherent Ising machine (CIM) [18]. In the Supplementary Materials section S1.4, we show that the polynomial PM model with arbitrary real-valued prefactors can be rescaled to the form:

$$\frac{dx_i}{dt} = -x_i - x_i |x_i|^{n-1} + \gamma (x_i^*)^{q-1} + \beta \sum_j^N J_{ij} x_j. \quad (7)$$

Here, $x_i \in \mathbb{C}$ is the complex-valued amplitude of spin i , β is the coupling strength, γ is the strength of phase discretization, n is the nonlinearity order, and q is the number of Potts states.

We propose annealing the model by increasing β and γ linearly over the course of the annealing with rates $\varepsilon_\beta = d\beta/dt$ and $\varepsilon_\gamma = d\gamma/dt = f_\gamma \varepsilon_\beta$, respectively, where f_γ is a constant factor. We show that the origin becomes unstable when β crosses the threshold $\beta_{th} = \mu_{max}^{-1}$, where μ_{max} is the principal eigenvalue of the coupling matrix (see Supplementary Materials section S2.1.1 for the stability analysis). Thus, the spins can be initialized at the origin, and β and γ can be increased arbitrarily slowly to ensure adiabatic annealing. The initial values for β and γ are set to $\beta_0 = \beta_{th}$ and $\gamma_0 = f_\gamma \beta_{th}$, respectively, so that the relative annealing rate $f_\gamma = \varepsilon_\gamma / \varepsilon_\beta$ can be controlled. For our studies, we set ε_β to 10^{-3} and sweep f_γ to find the optimal relative annealing rate for each problem instance (see Supplementary Materials section S7).

2.5 Sigmoid Potts machine model (Sigmoid PM) - this work

The sigmoid PM model is a new mathematical model proposed in this work, motivated by the observation that sigmoid nonlinearities enforce amplitude homogeneity and improve performance in analog IMs [18]. Unlike the other models considered here, the sigmoid PM is not directly connected to a specific hardware implementation. For Potts machines with complex-valued spins, simply replacing the polynomial nonlinearity with a sigmoid nonlinearity would not only saturate amplitudes, but also saturate the phases, which is undesirable. Instead, we propose a model that applies a sigmoid nonlinearity to the amplitude of the spins, while keeping the phase dynamics unaltered, enabled by using separate equations for amplitude and phase evolution (see Supplementary Materials section S1.5). The governing equations of the sigmoid PM model are then:

$$\frac{dr_i}{dt} = -r_i + \tanh\left(\alpha r_i + \gamma r_i^{q-1} \cos(q\theta_i) + \beta \sum_{j=1}^N J_{ij} r_j \cos(\theta_j - \theta_i)\right), \quad (8a)$$

$$\frac{d\theta_i}{dt} = -\gamma r_i^{q-2} \sin(q\theta_i) + \frac{\beta}{r_i} \sum_{j=1}^N J_{ij} r_j \sin(\theta_j - \theta_i). \quad (8b)$$

Here, $r_i \in \mathbb{R}$ is the amplitude of spin i , $\theta_i \in \mathbb{R}$ is the phase of spin i , α is the amplitude gain, β is the coupling strength, γ is the phase discretization strength, and q is the number of Potts states.

The sigmoid PM model can be annealed, either by increasing α or β , or both, along with γ . We choose to treat α as a constant and increase β and γ linearly over the course of the annealing with rates $\varepsilon_\beta = d\beta/dt$ and $\varepsilon_\gamma = d\gamma/dt = f_\gamma \varepsilon_\beta$, respectively, where f_γ is a constant

factor. Within this annealing scheme, lowering the constant α in practice slows the evolution of the system, which can support adiabatic annealing. The origin becomes unstable when β crosses the threshold $\beta_{th} = (1 - \alpha)\mu_{max}^{-1}$ or when α crosses the threshold $\alpha_{th} = 1 - \beta\mu_{max}$, where μ_{max} is the principal eigenvalue of the coupling matrix (see Supplementary Materials section S2.1.2 for the stability analysis). Thus, the spins can be initialized at the origin, and β and γ can be increased arbitrarily slowly to ensure adiabatic annealing. The initial values for β and γ are set to $\beta_0 = \beta_{th}$ and $\gamma_0 = f_\gamma \beta_{th}$, respectively. For our studies, we set ε_β to 10^{-3} and sweep f_γ as well as α for each problem instance (see Supplementary Materials section S7).

2.6 Sigmoid Ising machine with spin sign method (Sigmoid IM)

The sigmoid-based Ising machine (IM) is based on the coherent Ising machine (CIM) with sigmoid nonlinearity proposed by Böhm et al. [18]. Given an external bias field h_i for each spin i , and using the spin sign method recently been shown to be the most effective method for encoding external fields [26], the governing equations of the sigmoid IM becomes

$$\frac{dx_i}{dt} = -x_i + \tanh(\alpha x_i + \beta I_i) \quad (9a)$$

$$I_i = h_i + \sum_{j=1}^N J_{ij} \text{sgn}(x_j) \quad (9b)$$

where $x_i \in \mathbb{R}$ is the spin value of spin i , α is the amplitude gain, and β is the coupling strength, and I_i is the local field experienced by spin i . The local field is the sum of the coupling contributions from all other spins, weighted by their sign, and the external bias field h_i .

To represent Max- k -Cut problems, we use the Ising graph coloring formulation [7], which allows us to encode the target problem in the form of the coupling coefficients J_{ij} and the external bias field h_i .

The annealing scheme of this model entails increasing the coupling strength β linearly over the course of the annealing with a rate $\varepsilon_\beta = d\beta/dt$, while keeping α constant. As for the sigmoid PM, lowering the constant α slows the evolution of the system, which can support adiabatic annealing.

3 Methods

3.1 Numerical methods

We use the Euler–Maruyama method [28] to numerically integrate the governing equations of the Potts- and Ising-machine models. This method is a stochastic extension of the Euler method for solving ordinary differential equations and is suitable for simulating continuous-time dynamical systems with noise. See the Supplementary Materials

Table 2: GSet graphs used for benchmarking.

Graph	Vertices	Edges	Optimum Max-3-Cut	Optimum Max-4-Cut
G1	800	19176	15165	16803
G2	800	19176	15172	16809
G3	800	19176	15173	16806
G4	800	19176	15184	16814
G5	800	19176	15193	16816

The optimum cut values for the graphs interpreted as Max-3-Cut and Max-4-Cut problems are found in the literature [30].

section S5 for a more detailed description of the method and its implementation.

For all simulations, the noise factor of the stochastic process is set to $f_\xi = 10^{-4}$ to model a scenario with low experimental noise. The choice of time step size is critical for capturing the dynamics of the system accurately, especially for models with fast dynamics. The annealing of model parameters is implemented through the time dependency of $\mathbf{F}(\mathbf{x}, t)$. The time step size is chosen to ensure numerical stability and accuracy, in the range of 10^{-3} to 10^{-2} , depending on the model. See the Supplementary Materials section S7 for details on the choice of time step size for each model.

3.2 Problem selection and encoding

We use graphs from the GSet collection [29] as our benchmark set, which contains undirected graphs with edge weights of +1 or -1. For unweighted Max- k -Cut, we select only the +1-weighted instances. Table 2 lists the chosen graphs (names, vertex and edge counts) along with the optimum Max-3-Cut and Max-4-Cut values as reported in the literature, obtained via a multiple operator heuristic (MOH) [30].

The Max- k -Cut problems are encoded as Potts Hamiltonians by having one Potts spin with $q = k$ states represent each vertex and setting coupling coefficients $J_{ij} = J_{ki} = -1$ for edges with weight +1 between vertices i and k , and 0 otherwise. For the Ising machine, we use the Ising graph coloring formulation [7], which uses k Ising spins to represent each vertex (using one-hot encoding to represent the k states) and encodes the target problem in the form of the coupling coefficients J_{ij} and the external bias field h_i .

3.3 Performance metrics

The performance of the models is evaluated based on the optimality gap, which is defined as the difference between the optimum cut value of the problem and the maximum cut value found by the model, and can be normalized by

the optimum cut value for a relative optimality gap that can be used for comparison across different problems:

$$\text{Relative optimality gap} = \frac{C_{\text{opt}} - C_{\text{model}}}{C_{\text{opt}}}, \quad (10)$$

where C_{model} is the maximum cut value found by the model, and C_{opt} is the optimum cut value of the problem, either the global optimum or the best known solution for the problem.

Success rate, the fraction of runs that achieve the optimum cut value, is often used as a performance metric for combinatorial optimization problem solvers. However, the success rates are often zero for the studied models across the benchmark problems, which is why we instead use the relative optimality gap as a performance metric.

Time-to-solution is another popular performance metric, which is defined as the time taken by the model to reach a solution with a relative optimality gap below a certain threshold. This metric is not used in this work, as the models are optimized for maximum cut value rather than speed. Future comparative studies could include time-to-solution analysis to provide a more comprehensive performance evaluation.

3.4 Benchmarking procedure

Each model m is evaluated on each graph g by sweeping its hyperparameters h . For each triplet (m, g, h) , we run n_{runs} independent simulations of n_{steps} time steps, record the maximum cut value per run, and compute the relative optimality gap (Eq. (10)).

The number of runs n_{runs} is set to 100 for each hyperparameter combination, which is sufficient to obtain statistically significant results while keeping the computational cost manageable. The number of time steps n_{steps} depends on the model and is chosen to ensure that the system has enough time to converge. See the Supplementary Materials for details on the choice of n_{steps} for each model.

4 Qualitative study of model dynamics

In this section, we qualitatively study the dynamics of the five Potts machine (PM) models and one Ising machine (IM) model presented in the previous section. The goal of this qualitative study is to identify three key dynamical properties that characterize the evolution of the system. We will later show that these properties correlate with the performance of the models in solving Max- k -Cut. The dynamical properties are defined in the next section and are evaluated for specific models in the subsequent sections.

4.1 Selected dynamical properties for analysis

We focus on the following three dynamical properties:

Connected origin-to-solution branch requires that the trivial fixed point at the origin is initially stable and bifur-

cates into a nontrivial solution branch that evolves continuously from the origin.

Enforced amplitude homogeneity requires a mechanism that drives all spin amplitudes to the same value over time.

Enforced phase discretization requires that the continuous spin phases converge to the discrete set of phases corresponding to the Potts model states over time.

uses real-valued variables, phase discretization is not applicable.

4.2 Evaluation of dynamical properties

To evaluate the dynamical properties of the Potts machine models, we analyze their governing equations under the annealing schemes specified in Table 1. We highlight the key observations here and refer to the Supplementary Materials section S3 for an in-depth evaluation. Table 3 summarizes the evaluation for each model.

Figure 1 shows the time traces from numerical simulation of the five PM models and the IM model on a sample Max-3-Cut problem, showing how the dynamical properties manifest in the evolution of the system. The numerical simulations are described in section 3.1 with default parameters specified in the Supplementary Materials section S7.

The NEC model possesses a connected origin-to-solution branch through gain-controlled bifurcation, enforces amplitude homogeneity via dynamic gain adjustment, and achieves complete phase discretization through increasing coupling strength.

The q -PDC model lacks a connected origin-to-solution branch because the origin remains stable throughout annealing, and it does not enforce amplitude homogeneity. Amplitude seeding and rapid increase of γ can nevertheless escape the origin and yield complete phase discretization.

The q -SHIL model has intrinsically fixed and homogeneous amplitudes and enforces complete phase discretization. The connected-branch requirement is not applicable because the model has no origin fixed point.

The polynomial PM model exhibits a connected origin-to-solution branch but does not enforce amplitude homogeneity. Phase discretization depends on the annealing scheme; with nonlinear variation of γ versus β , e.g. $\gamma = f_\gamma \beta^2$, the requirement can be satisfied. However, with linear variation of γ versus β , as used in this work, the requirement is not satisfied.

The sigmoid PM model shows a connected origin-to-solution branch and enforces amplitude homogeneity. However, similarly to the polynomial PM model, complete phase discretization depends on the annealing scheme, and with linear variation of γ versus β , as used in this work, the requirement is not satisfied.

The sigmoid IM model exhibits a connected origin-to-solution branch and amplitude homogeneity due to the saturation of the sigmoid nonlinearity. Since the model

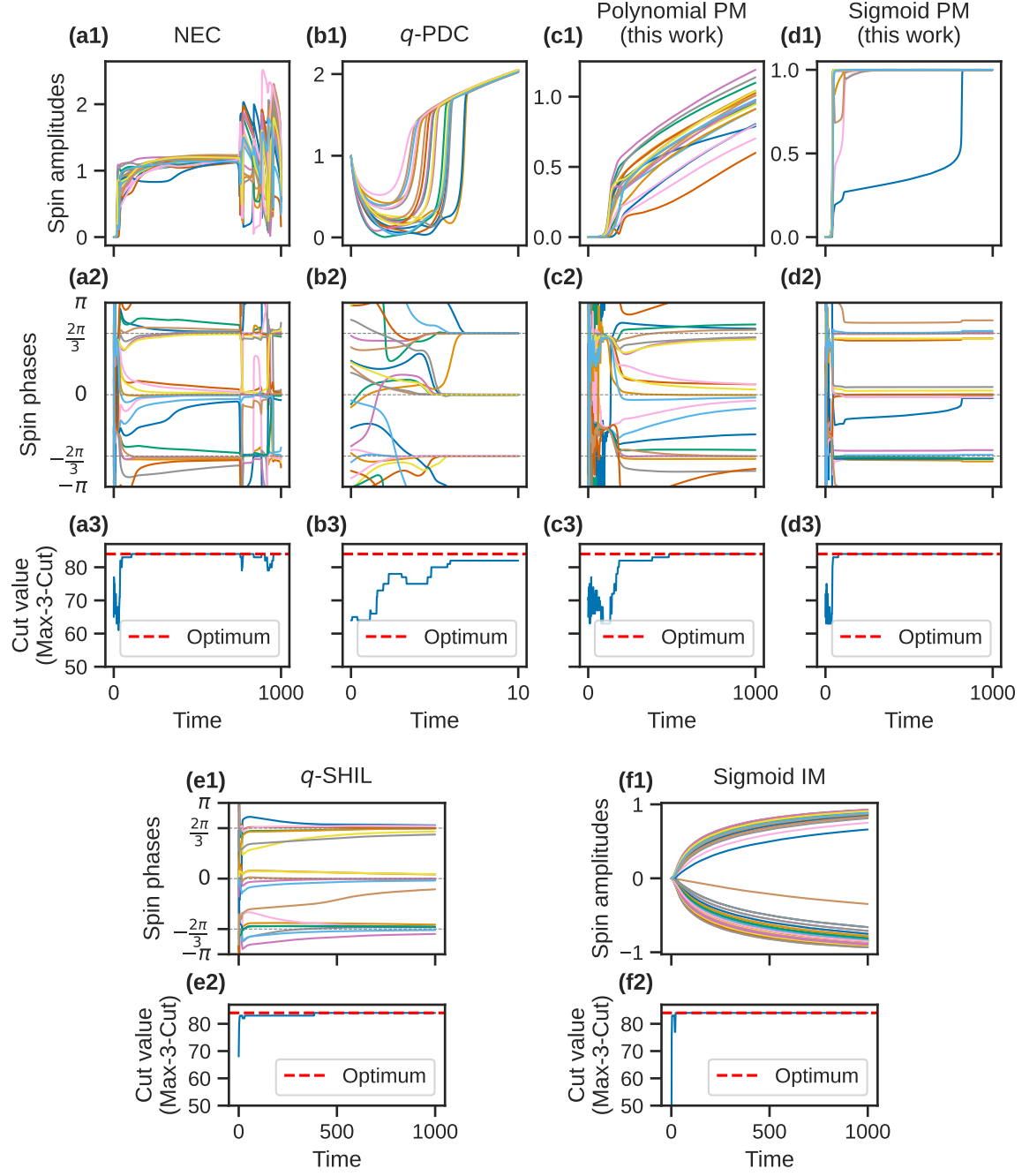


Figure 1: Time traces for each studied model on 20-vertex Max-3-Cut problem with 50% edge probability, generated using the Rudy graph generator [31] (see Supplementary Materials section S6 for the graph specification). We visualize the evolution of the amplitudes (a1 - d1, f1), phases (a2 - d2, e1), and the cut value (a3 - d3, e2 - f2) of the system. For the amplitude and phase plots, curves of different spins are shown in different colors, while the gray dashed lines indicate the discrete Potts phases. For cut value plots, the red dashed line indicates the optimum cut value of the problem. Default parameters listed in the Supplementary Materials section S7 are used for each model.

Table 3: Summary of dynamical properties for each model.

	NEC	q -PDC	q -SHIL	Polynomial PM	Sigmoid PM	Sigmoid IM
Connected origin-to-solution branch	✓	✗	NA	✓	✓	✓
Enforced amplitude homogeneity	✓	✗	✓	✗	✓	✓
Complete phase discretization	✓	✓	✓	(✓)	✗	NA

The evaluation of these properties is detailed in the Supplementary Materials section S3. A checkmark, ✓, indicates that the model satisfies the requirement, while a cross, ✗, indicates that it does not. The parenthesized checkmark, (✓), indicates that the requirement can be satisfied under an alternative annealing scheme. NA indicates that the requirement is not applicable to the model, e.g. because it does not have phases.

5 Benchmarking model performance

In this section, we perform a quantitative study of the performance of the studied Potts- and Ising machine models in solving combinatorial optimization problems. We use a benchmark set of selected graphs from the GSet collection [29], which we interpret as Max-3-Cut and Max-4-Cut problems. The problem selection and encoding is detailed in section 3.2.

The performance of the models is evaluated based on the relative optimality gap, which is defined as the difference between the optimum cut value of the problem and the maximum cut value found by the model, normalized by the optimum cut value (see section 3.3). Over a set of hyperparameter combinations, a number of simulations are run for each model, and the relative optimality gap is calculated for each run for later statistical analysis (see section 3.3).

Figure 2 shows the relative optimality gap distributions for Max-3-Cut and Max-4-Cut problems, across all models and benchmark graphs, with hyperparameters optimized for minimum mean optimality gap. Supplementary Materials section S8 provides more detailed results, including heatmaps of the hyperparameter tuning sweeps for each model and benchmark graph. For more details on the simulations settings and hyperparameter tuning sweeps, refer to the Supplementary Materials section S7.

The results show that the sigmoid IM model consistently achieves the smallest optimality gaps across all benchmark graphs, with median gaps around 0.1% for Max-3-Cut and around 0.2% for Max-4-Cut problems. The NEC model, which exhibits all three identified dynamical properties, follows closely behind the sigmoid IM in performance for Max-3-Cut, achieving median gaps below 0.2%, but the median gap increases to around 3.3% for Max-4-Cut. The polynomial PM and sigmoid PM models show competitive performance compared to other PMs with median gaps around 0.7% for Max-3-Cut and around 3.5% for Max-4-Cut. The q -SHIL performs slightly worse than the polynomial and sigmoid PMs, with median gaps around 1.2% for Max-3-Cut and around 4.5% for Max-4-Cut. The q -PDC model trails with higher gaps of around 4% for Max-3-Cut and around 6% for Max-4-Cut.

Note that in going from Max-3-Cut to Max-4-Cut, all Potts machine models show notably degraded performance, with median gaps increasing to $x\%$ for the best performers. Interestingly, the sigmoid IM maintains strong performance across both problem types. This will be discussed in more detail in section 6.4. It is also important to note that our analysis focuses on solution quality rather than convergence speed, and the relative performance of these models might differ when evaluated using time-to-solution or other metrics.

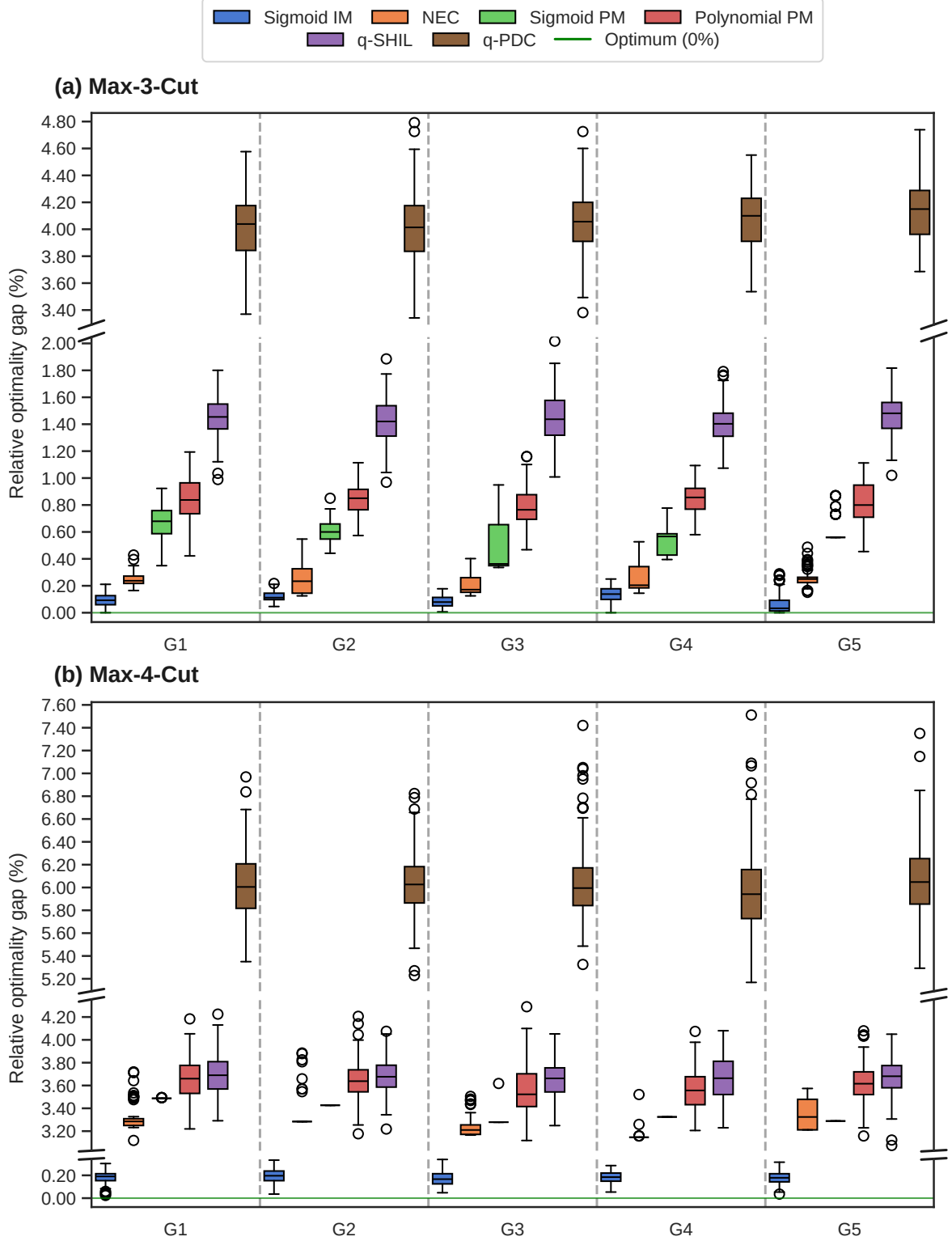


Figure 2: Relative optimality gap distributions for each model on (a) Max-3-Cut problems and (b) Max-4-Cut problems across GSet benchmark graphs G1-G5. Each model was evaluated using hyperparameters optimized for minimum mean optimality gap. Box plots show the distribution of relative optimality gaps (%) across 100 independent runs per graph-model combination. Lower values indicate better performance, with 0% representing the optimal solution (green horizontal line). The sigmoid IM consistently achieves the smallest optimality gaps, followed by the NEC model among Potts machines. See Supplementary Materials section S8 for detailed results, including hyperparameter tuning heatmaps for each model and graph.

6 Discussion

6.1 Relevance of identified dynamical properties

The performance of a physical solver, such as a Potts machine (PM) or an Ising machine (IM), is determined by its system dynamics, which are governed by its underlying equations and operational parameters, such as the annealing schedule. Our analysis reveals a strong correlation between the three identified dynamical properties and the solver’s performance on Max- k -Cut problems.

Our benchmarking results, measured by the relative optimality gap, show that models satisfying more of these requirements tend to perform better. The NEC model, which satisfies all three requirements, consistently outperformed the other PMs. The polynomial and sigmoid PMs, each satisfying two of the three requirements, also demonstrated competitive performance. Conversely, the q -PDC model, which satisfies only one requirement, performed poorly. The q -SHIL model fulfills the two applicable requirements but lacks the connected origin-to-solution branch, as discussed in section 6.2. These findings strongly suggest that the identified dynamical properties are crucial for the performance of PMs in solving combinatorial optimization problems.

The identified requirements—connected origin-to-solution branch, enforced amplitude homogeneity, and complete phase discretization—are not only relevant for the studied PM models but also provide a framework for evaluating and designing PMs in general. They can serve as a checklist for assessing new proposals in the literature, guiding hardware design choices to focus optimization efforts on the most impactful characteristics, and acting as explicit design objectives for novel architectures.

Our findings further indicate the promise of transitioning from the hardware-first approach to designing physical solvers, prioritizing implementation feasibility, to a dynamics-first approach incorporating dynamical systems analysis in order to identify promising architectures and guide parameter selection toward optimal performance.

6.2 Impact of origin-to-solution branch

The superior performance of the sigmoid PM over the q -SHIL model, despite both satisfying two dynamical properties, highlights that a connected origin-to-solution branch can be particularly important. This connection allows the system to traverse a low-amplitude regime where the interplay between stochastic noise and the deterministic drift term of the governing equations is pronounced, facilitating a more effective exploration of the energy landscape. In contrast, the q -SHIL model, which lacks amplitude dynamics and therefore this connection, operates in a fixed high-amplitude regime where the dynamics are dominated by the drift term, hindering landscape exploration. This suggests that the performance of fixed-amplitude models like q -SHIL might be improved by incorporating a noise annealing schedule, analogous to simulated annealing [32],

to emulate the exploratory benefits provided by an origin-connected solution branch.

6.3 Underperformance of Potts machine models compared to Ising machine model

A key finding of this study is the general underperformance of the studied PMs compared to the benchmark IM when evaluated using solution quality metrics. This result is surprising, as PMs are specifically formulated to solve multi-state problems such as Max- k -Cut. The IM, conversely, requires a problem mapping that introduces significant overhead in the number of spins and couplings to represent the multi-state variables [7]. This overhead is generally expected to increase the ruggedness of the energy landscape and degrade performance. Despite this, the IM model consistently outperforms the PM models across the benchmark problems in terms of optimality gap. However, it is important to note that this comparison is based solely on solution quality, and the ranking could be different when considering other metrics such as time-to-solution or energy efficiency, which we have not studied here.

The underperformance of the PM models could be due to several factors and warrants further study. It is important to note that significant research efforts have been dedicated to identifying high-performance IMs. The IM model used in this work [26] represents one of the most effective IMs in the literature. In contrast, the study of PMs remains in its early stages, so the most promising architectures and implementations of PMs may yet be undiscovered. This is exactly the motivation for this work, and highlights the need for further research into PMs to identify architectures and implementations that can match or exceed the performance of IMs. Additionally, while IMs may achieve better solution quality in our benchmarks, PMs could potentially offer advantages in terms of convergence speed, hardware implementation efficiency, or energy consumption that were not evaluated in this study.

6.4 Decreasing performance of Potts machine models going from Max-3-Cut to Max-4-Cut

Our benchmarks reveal a significant performance degradation for all PM models when transitioning from Max-3-Cut to Max-4-Cut problems, a trend not observed in the IM. This is especially surprising, as the representational overhead for an IM increases with k , while it remains constant for a PM. This discrepancy points to a fundamental mismatch between the objective function optimized by the studied PMs and the *standard Potts model* that correctly represents Max- k -Cut. The governing equations of the studied PMs, lead to a Lyapunov function (i.e. energy landscape) with an coupling term proportional to $-\sum J_{ij} \cos(\theta_i - \theta_j)$, depending on the angular separation between spin phases θ_i and θ_j . This is, in fact, the Hamiltonian of the *planar Potts model* [17], which differs from the standard Potts model Hamiltonian,

$-\sum J_{ij}\delta(s_i, s_j)$, that depends only on the discrete Potts states s_i and s_j .

While the Max- k -Cut problem directly maps to the standard Potts Hamiltonian with k states, it does not map directly to the planar Potts Hamiltonian for $k \geq 4$, as we will explain in the following. The Max- k -Cut cut value and the standard Potts Hamiltonian are defined in terms of the number of edges between pairs of spins in different states. The planar Potts Hamiltonian, on the other hand, is defined in terms of the angular separation between pairs of spins. The angular separation will be the same for all pairs of unique spins for $k \leq 3$, but not for $k \geq 4$, where the angular separation between pairs of spins depends on which pair is considered, e.g. whether they are adjacent, next-nearest neighbors, and so on. Therefore, for Max- k -Cut problems with $k \geq 4$, the considered PM models try not just to maximize the number of edges between pairs of spins in different states, but also to maximize the angular separation between pairs connected spins. This is an additional constraint not present in the included IM model, given its one-hot encoding of the problems [7]. The IM model therefore implements the standard Potts Hamiltonian. The additional constraint of angular separation imposed by the PM models leads to a mismatch between the energy landscape of the PMs and the target problem for Max-4-Cut, which is likely the reason for the observed performance degradation for Max-4-Cut compared to Max-3-Cut, where the angular separation constraint doesn't lead to a mismatch.

This observation highlights the importance of carefully considering the governing equations of PMs and their implications for the energy landscape when designing PMs for solving combinatorial optimization problems. It suggests that future research into PMs should focus on developing PM implementations with governing equations that do not impose unnecessary constraints on the target problems, such as the angular separation constraint found in this work.

7 Conclusion

In this work, we have presented the first systematic comparison between analog Potts machines (PMs) and an analog Ising machine (IM) for solving Max-3-Cut and Max-4-Cut problems. We studied three PM models from the literature: nonequilibrium condensate (NEC), q -photon downconversion (q -PDC), and q -th sub-harmonic injection locking (q -SHIL). In addition, we introduced two new variants, the polynomial PM and the sigmoid PM. An analog IM with sigmoid nonlinearity and spin-sign encoding was included as a reference. Through a combination of dynamical-systems analysis and numerical benchmarking on GSet problems, we identified three key dynamical properties that govern performance: a connected origin-to-solution branch, enforced amplitude homogeneity, and complete phase discretization.

Our benchmarks on Max-3-Cut and Max-4-Cut instances reveal that, despite their native representation of multi-state variables, all PMs exhibited larger mean and minimum optimality gaps than the reference IM, with the NEC model achieving the best performance among PMs. Both the polynomial and sigmoid PMs, which satisfy around two of the three dynamical properties, achieved competitive but still inferior solution quality relative to the NEC model, which satisfies all three requirements. The q -PDC and q -SHIL models, crucially lacking a connected origin-to-solution branch, showed the poorest performance. Notably, all PMs suffered a pronounced performance drop when transitioning from Max-3-Cut to Max-4-Cut, whereas the IM's performance remained stable. We attributed this decline to an unintentional angular separation constraint arising from the planar Potts Hamiltonian implemented by the PM dynamics, which mismatches the standard Potts model for Max-4-Cut and in general for Max- k -Cut with $k > 3$.

These findings lead to two main insights. First, solvers that exhibit a greater number of the selected dynamical properties reliably achieve higher solution quality. These requirements can therefore be used as normative standards guiding the design of PMs. Second, the considered PM formulations require fundamental revision to implement the standard Potts Hamiltonian for state counts larger than three.

As this work only studies the performance of PMs in terms of solution quality, future work should explore the performance of these models using other metrics, such as time-to-solution.

8 Data availability

The code used for the numerical simulations and the benchmarking procedure as well as the resulting data is available from the authors upon reasonable request. The benchmark set of graphs is publicly available from the GSet collection [29] and the best known cut values for the Max-3-Cut and Max-4-Cut problems are reported in the literature [30].

9 Author contributions

B.A.F performed the simulations and wrote the manuscript. R.D.P and P.B. supervised the project and provided critical feedback on the manuscript. All authors reviewed and approved the final manuscript.

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Supplementary Materials

S1 Rescaling and deriving governing equations

S1.1 Non-Equilibrium Condensates (NEC) model

For the non-equilibrium condensate model, assuming coupling strengths are initialized to the target coupling coefficients ($\Upsilon_{ij} = J_{ij}$) and setting the exponent of the magnitude $|x_i|$ to $n - 1$, the single-spin dynamics equation reads [23]

$$\frac{dx_i}{dt} = \alpha_i x_i - \sigma_c |x_i|^{n-1} x_i - h_i (x_i^*)^{q-1} + \sum_j^N J_{ij} x_j \quad (\text{S1.1})$$

Note that in the original formulation, the exponent of $|x_i|$ is 2, corresponding to $n = 3$. However, we shall see in section S4 that higher n is required to ensure amplitude bounding when $q > 3$, such as for Max-4-Cut problems, where $q = 4$. Scaling the complex amplitude by $x'_i = \sigma_c^s x_i$, where s is currently unspecified, yields

$$\sigma_c^{-s} \frac{dx'_i}{dt} = \sigma_c^{-s} \alpha_i x'_i - \sigma_c^{1-3s} |x'_i|^2 x'_i - \sigma_c^{(1-q)s} h_i (x_i'^*)^{q-1} + \sigma_c^{-s} \sum_j^N J_{ij} x'_j \quad (\text{S1.2})$$

$$\Rightarrow \frac{dx'_i}{dt} = \alpha_i x'_i - \sigma_c^{1-2s} |x'_i|^{n-1} x'_i - \sigma_c^{(2-q)s} h_i (x_i'^*)^{q-1} + \sum_j^N J_{ij} x'_j \quad (\text{S1.3})$$

It is evident that setting $s = 1/2$ gives

$$\frac{dx'_i}{dt} = \alpha_i x'_i - |x'_i|^{n-1} x'_i - \sigma_c^{1-\frac{q}{2}} h_i (x_i'^*)^{q-1} + \sum_j^N J_{ij} x'_j \quad (\text{S1.4})$$

With the effective parameter $\gamma = \sigma_c^{1-\frac{q}{2}} h_i$, the equation now reads

$$\frac{dx'_i}{dt} = \alpha_i x'_i - |x'_i|^{n-1} x'_i - \gamma (x_i'^*)^{q-1} + \sum_j^N J_{ij} x'_j \quad (\text{S1.5})$$

S1.2 q -Photon Down-Conversion (q -PDC) model

In the q -PDC model, the differential equation governing the time-evolution of spin i reads [13]

$$\frac{dx_i}{dt} = -\gamma_i x_i - g_s x_i |x_i|^{2(q-1)} - g (x_i^*)^{q-1} + \sum_j^N J_{ij} x_j \quad (\text{S1.6})$$

Assuming all spins share a uniform dampening factor $\gamma_i = \gamma_d$, this corresponds to the polynomial model with

$$n = 2q - 1, \quad c_1 = \gamma_d, \quad c_2 = g_s, \quad c_3 = g, \quad c_4 = 1 \quad (\text{S1.7})$$

The equation can therefore be rescaled as (see eq. S1.19)

$$\frac{dx'_i}{dt'} = -x'_i - x'_i |x'_i|^{2(q-1)} - \gamma (x_i'^*)^{q-1} + \beta \sum_j^N J_{ij} x'_j, \quad (\text{S1.8})$$

$$t' = \gamma_d t, \quad x'_i = \sqrt{\frac{g_s}{\gamma_d}} x_i, \quad \gamma = \frac{g}{\gamma_d} \left(\frac{g_s}{\gamma_d} \right)^{\frac{2-q}{n-1}}, \quad \beta = \frac{1}{\gamma_d} \quad (\text{S1.9})$$

S1.3 q^{th} -harmonic Subharmonic Injection Locking (q -SHIL) model

In the q^{th} -harmonic subharmonic injection locking model, the phase dynamics for oscillator i is described by the differential equation

$$\frac{d\theta_i}{dt} = K_s \sin(q\theta_i) + K \sum_j^N J_{ij} \sin(\theta_j - \theta_i) \quad (\text{S1.10})$$

Assuming $K \neq 0$, rescaling the time by $t' = Kt$ results in

$$\frac{d\theta_i}{dt'} = \frac{K_s}{K} \sin(q\theta_i) + \sum_j^N J_{ij} \sin(\theta_j - \theta_i) \quad (\text{S1.11})$$

By introducing the effective parameter $\gamma = K_s/K$, the equation reads

$$\frac{d\theta_i}{dt'} = \gamma \sin(q\theta_i) + \sum_j^N J_{ij} \sin(\theta_j - \theta_i) \quad (\text{S1.12})$$

S1.4 Polynomial Potts machine model

In the polynomial model, the differential equation governing the time-evolution of spin i reads

$$\frac{dx_i}{dt} = -c_1 x_i - c_2 x_i |x_i|^{n-1} - c_3 (x_i^*)^{q-1} + c_4 \sum_j^N J_{ij} x_j, \quad x_i \in \mathbb{C}, \quad c_1, c_2, c_3, c_4 \in \mathbb{R} \setminus \{0\} \quad (\text{S1.13})$$

Scaling the time by $t' = c_1 t$ results in

$$\frac{dx_i}{dt'} = -x_i - \frac{c_2}{c_1} x_i |x_i|^{n-1} - \frac{c_3}{c_1} (x_i^*)^{q-1} + \frac{c_4}{c_1} \sum_j^N J_{ij} x_j \quad (\text{S1.14})$$

Scaling the spin complex amplitude $x'_i = s x_i$ where s is currently unspecified gives

$$s^{-1} \frac{dx'_i}{dt'} = -s^{-1} x'_i - \frac{c_2}{c_1} s^{-n} x'_i |x'_i|^{n-1} - \frac{c_3}{c_1} s^{-(q-1)} (x_i^*)^{q-1} + \frac{c_4}{c_1} s^{-1} \sum_j^N J_{ij} x'_j \quad (\text{S1.15})$$

$$\Rightarrow \frac{dx'_i}{dt'} = -x'_i - \frac{c_2}{c_1} s^{1-n} x'_i |x'_i|^{n-1} - \frac{c_3}{c_1} s^{2-q} (x_i^*)^{q-1} + \frac{c_4}{c_1} \sum_j^N J_{ij} x'_j \quad (\text{S1.16})$$

It is now evident that setting $s = \left(\frac{c_2}{c_1}\right)^{\frac{1}{n-1}}$ such that $s^{1-n} = \left(\frac{c_2}{c_1}\right)^{-1}$ yields

$$\frac{dx'_i}{dt'} = -x'_i - x'_i |x'_i|^{n-1} - \frac{c_3}{c_1} \left(\frac{c_2}{c_1}\right)^{\frac{2-q}{n-1}} (x_i^*)^{q-1} + \frac{c_4}{c_1} \sum_j^N J_{ij} x'_j \quad (\text{S1.17})$$

Now, the following two effective parameters are introduced

$$\gamma = \frac{c_3}{c_1} \left(\frac{c_2}{c_1}\right)^{\frac{2-q}{n-1}}, \quad \beta = \frac{c_4}{c_1}, \quad (\text{S1.18})$$

such that γ is the prefactor on the phase-discretizing term of the equation whereas β is the prefactor on the coupling term. The equation then reads

$$\frac{dx'_i}{dt'} = -x'_i - x'_i |x'_i|^{n-1} - \gamma (x_i^*)^{q-1} + \beta \sum_j^N J_{ij} x'_j \quad (\text{S1.19})$$

In models that anneal by decreasing c_1 , this annealing corresponds to increasing β and having γ follow β via the expression

$$\gamma(\beta) = \frac{c_3}{c_1} \left(\frac{c_2}{c_1}\right)^{\frac{2-q}{n-1}} = \frac{\beta c_3}{c_4} \left(\frac{\beta c_2}{c_4}\right)^{\frac{2-q}{n-1}} = \frac{c_3}{c_4} \left(\frac{c_2}{c_4}\right)^{\frac{2-q}{n-1}} \beta^{1+\frac{2-q}{n-1}} = f_\gamma \beta^{1+\frac{2-q}{n-1}} \quad (\text{S1.20})$$

Here, f_γ is the γ -factor given by

$$f_\gamma = \frac{c_3}{c_4} \left(\frac{c_2}{c_4}\right)^{\frac{2-q}{n-1}} \quad (\text{S1.21})$$

S1.5 Sigmoid Potts machine model

To derive the sigmoid Potts machine model presented in the main text section 2.5, we start from the rescaled polynomial model equation given in eq. S1.19, rewrite it in terms of separate the amplitude and phase equations, and then apply the sigmoid function to the amplitude equation only.

For the rescaled polynomial model, the differential equation for spin i is

$$\frac{dx}{dt} = -x_i - x_i |x_i|^{n-1} + \gamma (x_i^*)^{q-1} + \beta \sum_j^N J_{ij} x_j \quad (\text{S1.22})$$

Expressing the complex amplitude by $x_i = r_i e^{i\theta_i}$, where r_i is its amplitude θ_i its phase, the left-hand side of the equation is expanded using the product rule:

$$\frac{dx}{dt} = e^{i\theta_i} \frac{d}{dt} r_i + r_i \frac{d}{dt} e^{i\theta_i} = e^{i\theta_i} \frac{dr_i}{dt} + i r_i e^{i\theta_i} \frac{d\theta_i}{dt} = e^{i\theta_i} \left(\frac{dr_i}{dt} + i r_i \frac{d\theta_i}{dt} \right) \quad (\text{S1.23})$$

When expressing x_i in terms of r_i and θ_i , the right-hand side of the spin dynamics equation yields

$$\frac{dx}{dt} = -r_i e^{i\theta_i} - e^{i\theta_i} |r_i|^n + \gamma r_i^{q-1} e^{-i(q-1)\theta_i} + \beta \sum_j^N J_{ij} r_j e^{i\theta_j} \quad (\text{S1.24})$$

Inserting the expanded left-hand side gives

$$e^{i\theta_i} \left(\frac{dr_i}{dt} + i r_i \frac{d\theta_i}{dt} \right) = -r_i e^{i\theta_i} - e^{i\theta_i} |r_i|^n + \gamma r_i^{q-1} e^{-i(q-1)\theta_i} + \beta \sum_j^N J_{ij} r_j e^{i\theta_j} \quad (\text{S1.25})$$

$$\Rightarrow \frac{dr_i}{dt} + i r_i \frac{d\theta_i}{dt} = -r_i - |r_i|^n + \gamma r_i^{q-1} e^{-iq\theta_i} + \beta \sum_j^N J_{ij} r_j e^{i(\theta_j - \theta_i)} \quad (\text{S1.26})$$

$$= -r_i - |r_i|^n + \gamma r_i^{q-1} [\cos(q\theta_i) - i \sin(q\theta_i)] + \beta \sum_j^N J_{ij} r_j [\cos(\theta_j - \theta_i) + i \sin(\theta_j - \theta_i)] \quad (\text{S1.27})$$

From this, we can separate the real and imaginary parts:

$$\frac{dr_i}{dt} = -r_i - |r_i|^n + \gamma r_i^{q-1} \cos(q\theta_i) + \beta \sum_j^N J_{ij} r_j \cos(\theta_j - \theta_i), \quad (\text{S1.28})$$

and

$$i r_i \frac{d\theta_i}{dt} = -i \gamma r_i^{q-1} \sin(q\theta_i) + i \beta \sum_j^N J_{ij} r_j \sin(\theta_j - \theta_i) \quad (\text{S1.29})$$

$$\Rightarrow \frac{d\theta_i}{dt} = -\gamma r_i^{q-2} \sin(q\theta_i) + \frac{\beta}{r_i} \sum_j^N J_{ij} r_j \sin(\theta_j - \theta_i) \quad (\text{S1.30})$$

Note that to divide by r_i in the phase equation, $r_i \neq 0$ must hold, meaning that for spins with zero amplitude, the phase is undefined.

By separating the amplitude and phase parts of the governing equation like this, the sigmoid saturation of the amplitudes can be introduced without affecting phase dynamics. Similarly to how the sigmoid Ising machine differs from the polynomial Ising machine [18], the polynomial amplitude equation is altered by placing the coupling term, phase-discretizing term and a linear term with prefactor α inside a hyperbolic tangent nonlinearity:

$$\frac{dr_i}{dt} = -r_i + \tanh \left(\alpha r_i + \gamma r_i^{q-1} \cos(q\theta_i) + \beta \sum_j^N J_{ij} r_j \cos(\theta_j - \theta_i) \right) \quad (\text{S1.31})$$

S2 Stability and fixed-point analysis

The stability of a given fixed point of a system of equations can be determined from the eigenvalues of the Jacobian matrix $\mathcal{U}$ at the fixed point. A fixed point $\mathbf{x}^*$ is defined as a point where the time-derivative of all variables is zero:

$$\left. \frac{d\mathbf{x}}{dt} \right|_{\mathbf{x}=\mathbf{x}^*} = \mathbf{0} \quad (\text{S2.32})$$

The Jacobian matrix is the matrix of all first-order derivatives of the system of equations

$$\mathcal{U} = \begin{bmatrix} \frac{\partial}{\partial x_1} \frac{dx_1}{dt} & \frac{\partial}{\partial x_2} \frac{dx_1}{dt} & \cdots & \frac{\partial}{\partial x_N} \frac{dx_1}{dt} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial}{\partial x_1} \frac{dx_N}{dt} & \frac{\partial}{\partial x_2} \frac{dx_N}{dt} & \cdots & \frac{\partial}{\partial x_N} \frac{dx_N}{dt} \end{bmatrix}, \quad (\text{S2.33})$$

with elements

$$\mathcal{U}_{ij} = \frac{\partial}{\partial x_j} \frac{dx_i}{dt} \quad (\text{S2.34})$$

When evaluated at a fixed point, a negative eigenvalue indicates that the fixed point is attracting in the direction of the eigenvector corresponding to the eigenvalue. Vice versa, a positive eigenvalue indicates that the fixed point is repellent in the direction of the eigenvector corresponding to the eigenvalue. Therefore, when all eigenvalues are negative, the fixed point is attracting in all directions and is said to be stable. When the principal eigenvalue of the Jacobian matrix evaluated at a fixed point changes from negative to positive, that fixed point becomes unstable.

S2.1 Stability of the origin fixed point

Here, the stability of the origin fixed point $\mathbf{x}^* = \mathbf{0}$ is analyzed for the different Potts machine models presented in the main text, except for the q -SHIL model, which does not have amplitudes, and thus no origin. For all other Potts machine models, the origin is a fixed point models as $d\mathbf{x}/dt|_{\mathbf{x}=\mathbf{0}} = \mathbf{0}$.

S2.1.1 NEC, q -PDC, and polynomial Potts machine models

Both the NEC, q -PDC, and polynomial Potts machine governing equations can be written in the form of equation S1.19 with a linear term prefactor α_i :

$$\frac{dx_i}{dt} = \alpha_i x_i - x_i |x_i|^{n-1} - \gamma (x_i^*)^{q-1} + \beta \sum_l^N J_{il} x_l, \quad (\text{S2.35})$$

where for the NEC model, α_i is given by the dynamic gain equation (4b) and $\beta = 1$. For the q -PDC and polynomial Potts machine models, $\alpha_i = -1 \forall i$. In the following, the case where all spin-dependent gains are set to the same value $\alpha_i = \alpha \forall i$ is considered. Assuming $n > 1$ and $q > 2$, linearizing the system of equations around the origin ($|x_i| \ll 1$) by neglecting the terms of higher than first order in x_i gives

$$\frac{d\mathbf{x}}{dt} = \alpha \mathbf{x} + \beta \mathbf{J} \mathbf{x} = (\alpha \mathbf{I} + \beta \mathbf{J}) \mathbf{x} = \mathcal{U}|_{\mathbf{x}=\mathbf{0}} \mathbf{x}, \quad (\text{S2.36})$$

where $\mathbf{I}$ is the $N \times N$ identity matrix, $\mathbf{J}$ is the coupling matrix with elements J_{ij} , and $\mathcal{U}|_{\mathbf{x}=\mathbf{0}}$ is the Jacobian matrix evaluated at the origin fixed point. Eigenvalues of the Jacobian matrix at the origin fixed point are therefore

$$\lambda_k = \alpha + \beta \mu_k, \quad (\text{S2.37})$$

where μ_k is the k^{th} eigenvalue of the coupling matrix. The principal eigenvalue of the coupling matrix is denoted $\mu_{\max}$, such that given $\beta > 0$, the principal eigenvalue of the Jacobian matrix is

$$\lambda_{\max} = \alpha + \beta \mu_{\max}, \quad (\text{S2.38})$$

While $\lambda_{\max} < 0$, the origin is stable, as all other eigenvalues are smaller than the principal eigenvalue and thus also negative. The principal eigenvalue changes sign from negative to positive at the threshold given by

$$0 = \alpha_{\text{th}} + \beta \mu_{\max} \Rightarrow \alpha_{\text{th}} = -\beta \mu_{\max} \text{ or } \beta_{\text{th}} = -\frac{\alpha}{\mu_{\max}}, \quad (\text{S2.39})$$

depending on which of the two parameters, α or β , is varied during annealing. As the value of $\lambda_{\max}$ increases with α and β , the origin is therefore stable for $\alpha < \alpha_{\text{th}}$ (or $\beta < \beta_{\text{th}}$) and unstable for $\alpha > \alpha_{\text{th}}$ (or $\beta > \beta_{\text{th}}$). When the gain-modulating equation of the NEC model is introduced, the origin might not remain stable indefinitely as α will increase automatically while $r_{\text{th}} > 0$. Yet the origin is still attracting in all directions for $\alpha < \alpha_{\text{th}}$ and repellent in at least one direction for $\alpha > \alpha_{\text{th}}$.

S2.1.2 Sigmoid Potts machine model

The sigmoid model is governed by separate equations for the spin amplitude and spin phase:

$$\frac{dr_i}{dt} = -r_i + \tanh \left(\alpha r_i - \gamma r_i^{q-1} \cos(q\theta_i) + \beta \sum_j^N J_{ij} r_j \cos(\theta_j - \theta_i) \right) \quad (\text{S2.40})$$

$$\frac{d\theta_i}{dt} = \gamma r_i^{q-2} \sin(q\theta) + \frac{\beta}{r_i} \sum_j^N J_{ij} r_j \sin(\theta_j - \theta_i) \quad (\text{S2.41})$$

To simplify notation the following quantities are introduced:

$$R_i = \alpha r_i - \gamma r_i^{q-1} \cos(q\theta_i) + \beta \sum_j^N J_{ij} r_j \cos(\theta_j - \theta_i) \quad (\text{S2.42})$$

$$S_i = \frac{dr_i}{dt} + r_i = \tanh(R_i) \quad (\text{S2.43})$$

$$T_i = r_i \frac{d\theta_i}{dt} = \gamma r_i^{q-1} \sin(q\theta) + \beta \sum_j^N J_{ij} r_j \sin(\theta_j - \theta_i) \quad (\text{S2.44})$$

It is noted that at the origin $\mathbf{x} = \mathbf{r} = \mathbf{0}$, all these quantities are zero:

$$R_i|_{\mathbf{x}=\mathbf{0}} = 0, \quad S_i|_{\mathbf{x}=\mathbf{0}} = \tanh R_i|_{\mathbf{x}=\mathbf{0}} = 0, \quad T_i|_{\mathbf{x}=\mathbf{0}} = 0 \quad (\text{S2.45})$$

Cf. supplementary materials S1.5, the amplitude and phase equation can be combined to an equation for the complex amplitude by

$$\frac{dx}{dt} = e^{i\theta_i} \left(\frac{dr_i}{dt} + i r_i \frac{d\theta_i}{dt} \right) = e^{i\theta_i} (S_i - r_i + iT_i) = -x_i + e^{i\theta_i} (S_i + iT_i) \quad (\text{S2.46})$$

Element ij of the Jacobian matrix is therefore

$$\mathcal{U}_{ij} = \frac{\partial}{\partial x_j} \frac{dx}{dt} = -\delta_{ij} + \frac{\partial}{\partial x_j} e^{i\theta_i} S_i + \frac{\partial}{\partial x_j} i e^{i\theta_i} T_i \quad (\text{S2.47})$$

The term with S_i can be expanded with the product rule:

$$\frac{\partial}{\partial x_j} e^{i\theta_i} S_i = S_i \frac{\partial}{\partial x_j} e^{i\theta_i} + e^{i\theta_i} \frac{\partial}{\partial x_j} S_i \quad (\text{S2.48})$$

At the origin, only the second of these terms are non-zero due to S_i being zero at the origin:

$$\left. \frac{\partial}{\partial x_j} e^{i\theta_i} S_i \right|_{\mathbf{x}=\mathbf{0}} = S_i \left. \frac{\partial}{\partial x_j} e^{i\theta_i} \right|_{\mathbf{x}=\mathbf{0}} + e^{i\theta_i} \left. \frac{\partial}{\partial x_j} S_i \right|_{\mathbf{x}=\mathbf{0}} = e^{i\theta_i} \left. \frac{\partial}{\partial x_j} S_i \right|_{\mathbf{x}=\mathbf{0}} \quad (\text{S2.49})$$

The partial derivative of $S_i = \tanh R_i$ can be expanded using the chain rule:

$$\frac{\partial}{\partial x_j} S_i = \left(\frac{\partial}{\partial R_i} S_i \right) \left(\frac{\partial}{\partial x_j} R_i \right) \quad (\text{S2.50})$$

The first factor simply takes unit value at the origin:

$$\frac{\partial}{\partial R_i} S_i = \text{sech}^2 R_i, \quad \left. \frac{\partial}{\partial R_i} S_i \right|_{\mathbf{x}=\mathbf{0}} = 1 \quad (\text{S2.51})$$

From this follows that

$$\left. \frac{\partial}{\partial x_j} S_i \right|_{\mathbf{x}=\mathbf{0}} = \left. \frac{\partial}{\partial x_j} R_i \right|_{\mathbf{x}=\mathbf{0}} \quad (\text{S2.52})$$

and

$$\left. \frac{\partial}{\partial x_j} e^{i\theta_i} S_i \right|_{\mathbf{x}=\mathbf{0}} = e^{i\theta_i} \left. \frac{\partial}{\partial x_j} S_i \right|_{\mathbf{x}=\mathbf{0}} = e^{i\theta_i} \left. \frac{\partial}{\partial x_j} R_i \right|_{\mathbf{x}=\mathbf{0}} \quad (\text{S2.53})$$

Now, it is noted that

$$\left. \frac{\partial}{\partial x_j} e^{i\theta_i} R_i \right|_{\mathbf{x}=0} = R_i \left. \frac{\partial}{\partial x_j} e^{i\theta_i} \right|_{\mathbf{x}=0} + e^{i\theta_i} \left. \frac{\partial}{\partial x_j} R_i \right|_{\mathbf{x}=0} = e^{i\theta_i} \left. \frac{\partial}{\partial x_j} R_i \right|_{\mathbf{x}=0}, \quad (\text{S2.54})$$

such that the above is equivalent to

$$\left. \frac{\partial}{\partial x_j} e^{i\theta_i} S_i \right|_{\mathbf{x}=0} = \left. \frac{\partial}{\partial x_j} e^{i\theta_i} R_i \right|_{\mathbf{x}=0} \quad (\text{S2.55})$$

Turning our attention back to the Jacobian matrix, its elements at the origin are given by

$$\begin{aligned} \mathcal{U}_{ij} \Big|_{\mathbf{x}=0} &= -\delta_{ij} + \left. \frac{\partial}{\partial x_j} (e^{i\theta_i} S_i) \right|_{\mathbf{x}=0} + \left. \frac{\partial}{\partial x_j} (ie^{i\theta_i} T_i) \right|_{\mathbf{x}=0} \\ &= -\delta_{ij} + \left. \frac{\partial}{\partial x_j} (e^{i\theta_i} R_i) \right|_{\mathbf{x}=0} + \left. \frac{\partial}{\partial x_j} (ie^{i\theta_i} T_i) \right|_{\mathbf{x}=0} \end{aligned} \quad (\text{S2.56})$$

Combining the two last terms yields

$$\mathcal{U}_{ij} \Big|_{\mathbf{x}=0} = -\delta_{ij} + \left. \frac{\partial}{\partial x_j} e^{i\theta_i} (R_i + iT_i) \right|_{\mathbf{x}=0} \quad (\text{S2.57})$$

Most of the work in this derivation thus far was about showing that R_i can substitute S_i in the above expression. This was done as $e^{i\theta_i} (R_i + iT_i)$ is easily differentiable with respect to x_j , whereas $e^{i\theta_i} (S_i + iT_i)$ is not. To see this, first the sum $R_i + iT_i$ is calculated:

$$R_i + iT_i = \alpha r_i - \gamma r_i^{q-1} e^{-iq\theta_i} + \beta \sum_j^N J_{ij} r_j e^{i(\theta_j - \theta_i)} \quad (\text{S2.58})$$

Factoring in $e^{i\theta_i}$ yields an expression in terms of complex amplitudes instead of amplitudes and phases:

$$e^{i\theta_i} (R_i + iT_i) = \alpha x_i - \gamma r_i^{q-1} e^{-i(q-1)\theta_i} + \beta \sum_j^N J_{ij} r_j e^{i\theta_j} = \alpha x_i - \gamma (x_i^*)^{q-1} + \beta \sum_j^N J_{ij} x_j \quad (\text{S2.59})$$

Taking the partial derivative is now straightforward:

$$\left. \frac{\partial}{\partial x_j} e^{i\theta_i} (R_i + iT_i) \right|_{\mathbf{x}=0} = \alpha \delta_{ij} + \beta J_{ij} \quad (\text{S2.60})$$

The Jacobian matrix elements at the origin are therefore

$$\mathcal{U}_{ij} \Big|_{\mathbf{x}=0} = (\alpha - 1) \delta_{ij} + \beta J_{ij}, \quad (\text{S2.61})$$

corresponding to Jacobian matrix

$$\mathcal{U} \Big|_{\mathbf{x}=0} = (\alpha - 1) \mathbf{I} + \beta \mathbf{J} \quad (\text{S2.62})$$

with eigenvalues

$$\lambda_k = \alpha - 1 + \beta \mu_k \quad (\text{S2.63})$$

Assuming $\beta > 0$, the principal eigenvalue is

$$\lambda_{\max} = \alpha - 1 + \beta \mu_{\max} \quad (\text{S2.64})$$

The destabilization threshold where $\lambda_{\max} = 0$ can then either be expressed in terms of α or β :

$$\alpha_{\text{th}} = 1 - \beta \mu_{\max}, \quad \beta_{\text{th}} = \frac{1 - \alpha}{\mu_{\max}} \quad (\text{S2.65})$$

S2.2 Evolution of fixed points following origin destabilization

S2.2.1 NEC, q-PDC, and polynomial Potts machine models

For the NEC, q -PDC, and polynomial Potts machine models, the governing equations linearized around the origin (Eq. S2.36) take the form

$$\frac{d\mathbf{x}}{dt} = \mathcal{U} \Big|_{\mathbf{x}=0} \mathbf{x}, \quad (\text{S2.66})$$

where $\mathcal{U}|_{\mathbf{x}=0}$ is the Jacobian matrix evaluated at the origin, given by

$$\mathcal{U}|_{\mathbf{x}=0} = \alpha \mathbf{I} + \beta \mathbf{J}. \quad (\text{S2.67})$$

The first eigenvalue of the Jacobian matrix that crosses zero and destabilizes the origin is the principal eigenvalue $\lambda_{\max} = \alpha + \beta \mu_{\max}$, with corresponding eigenvector $\mathbf{v}_{\max}$. Therefore, when the origin becomes unstable, the system will evolve along the direction of $\mathbf{v}_{\max}$, such that

$$\mathbf{x}(t) = c(t) \mathbf{v}_{\max}, \quad (\text{S2.68})$$

where $c(t) \in \mathbb{C}$ is a time-dependent complex-valued coefficient. As the coupling matrix is real-valued and symmetric, the eigenvector $\mathbf{v}_{\max}$ can be chosen to be real-valued. Therefore, all spin complex amplitudes will initially evolve away from the origin either parallel (if $v_{\max,i} > 0$) or anti-parallel (if $v_{\max,i} < 0$) to the complex coefficient $c(t)$, such that the phases are $\theta_i \in \{\arg(c), \arg(c) + \pi\} \forall i$. We note that $\lambda_{\max} = \alpha + \beta \mu_{\max}$ and write $v_i = v_{\max,i}$ for brevity. To describe the evolution of $c(t)$ after the origin destabilization, the higher-order terms must be included in the governing equations:

$$\frac{dx_i}{dt} = \frac{dc}{dt} v_i = \lambda_{\max} c v_i - c v_i |c|^{n-1} |v_i|^{n-1} - \gamma (c^*)^{q-1} v_i^{q-1} \quad (\text{S2.69})$$

If we choose the principal eigenvector to be normalized such that $\|\mathbf{v}\|_2 = 1$, we can easily extract the time-evolution of the global complex coefficient $c(t)$:

$$\frac{dc}{dt} = \frac{dc}{dt} \sum_i v_i^2 = \sum_i v_i \frac{dc}{dt} v_i = \sum_i v_i \frac{dx_i}{dt} \quad (\text{S2.70})$$

$$= \lambda_{\max} c - C_n c |c|^{n-1} - \gamma K_q (c^*)^{q-1} \quad (\text{S2.71})$$

where

$$C_n = \sum_i |v_i|^{n+1}, \quad K_q = \sum_i v_i^q. \quad (\text{S2.72})$$

are constants that depend on the principal eigenvector and the nonlinearity order and number of Potts states, respectively. At a steady state, $dc/dt = 0$, and the equation for c becomes

$$(\lambda_{\max} - C_n |c|^{n-1}) c = \gamma K_q (c^*)^{q-1}. \quad (\text{S2.73})$$

Multiplying both sides by c^{q-1} gives

$$(\lambda_{\max} - C_n |c|^{n-1}) c^q = \gamma K_q |c|^{2(q-1)}. \quad (\text{S2.74})$$

The right-hand side is real-valued as $\gamma K_q \in \mathbb{R}$, and the left-hand side must therefore also be real-valued, meaning that c^q is real-valued and of the same sign as K_q , assuming $\gamma > 0$. Therefore, the phase of c must be one of the q discrete values given by

$$\arg(c) = \frac{2\pi k + \arg(K_q)}{q} = \frac{2\pi k + \delta}{q}, \quad k = 0, \dots, q-1, \quad \delta = \begin{cases} 0, & K_q > 0, \\ \pi, & K_q < 0. \end{cases} \quad (\text{S2.75})$$

corresponding to the q discrete phases of the planar Potts model in $[0, 2\pi]$ offset by π/q if $K_q < 0$.

The amplitude $|c|$ can be found by taking the absolute value of the steady-state equation and dividing by $|c| > 0$:

$$|\lambda_{\max} - C_n |c|^{n-1}| = \gamma |K_q| |c|^{q-2}. \quad (\text{S2.76})$$

In section S4, it is shown that $n \geq q$ is required to ensure amplitude bounding for all coupling matrices. Assuming $n \geq q$ and $q > 2$, the $|c|^{q-2}$ term will dominate close to the origin where $|c| \ll 1$, and leading order terms give the fixed-point amplitude coefficient as

$$|c| \approx \left(\frac{\lambda_{\max}}{\gamma |K_q|} \right)^{\frac{1}{q-2}}, \quad |c| \ll 1. \quad (\text{S2.77})$$

Thus, when α or β is increased beyond the origin destabilization threshold, the fixed points will initially evolve from the origin according to

$$\mathbf{x} \approx \left(\frac{\lambda_{\max}}{\gamma |K_q|} \right)^{\frac{1}{q-2}} e^{i \frac{2\pi k + \delta}{q}} \mathbf{v}_{\max}, \quad (\text{S2.78})$$

where

$$\lambda_{\max} = \alpha + \beta\mu_{\max} \ll 1, \quad (\text{S2.79})$$

$$k = 0, \dots, q-1, \quad (\text{S2.80})$$

$$K_q = \sum_i^N v_i^q, \quad (\text{S2.81})$$

$$\delta = \begin{cases} 0, & K_q > 0, \\ \pi, & K_q < 0, \end{cases} \quad (\text{S2.82})$$

and $\mathbf{v}_{\max}$ is the principal eigenvector of the coupling matrix $\mathbf{J}$ with eigenvalue $\mu_{\max}$.

The above shows that after the origin is destabilized, the system bifurcates and all spins initially evolve parallel or anti-parallel to one of these q discrete phases, depending on the sign of the corresponding element in the principal eigenvector. The amplitude grows as a power law in $\lambda_{\max}$ with exponent $1/(q-2)$ close to the origin. As $\lambda_{\max}$ is increased further from zero, the amplitude will deviate from this power law as higher-order terms become significant.

S2.2.2 Sigmoid Potts machine model

As found in section S2.1.2, for the sigmoid Potts machine model, the Jacobian matrix at the origin is given by

$$\mathcal{U}|_{\mathbf{x}=\mathbf{0}} = (\alpha - 1)\mathbf{I} + \beta\mathbf{J}, \quad (\text{S2.83})$$

with principal eigenvalue $\lambda_{\max} = \alpha - 1 + \beta\mu_{\max}$ and corresponding eigenvector $\mathbf{v}_{\max}$, which is also the principal eigenvector of the coupling matrix $\mathbf{J}$ with eigenvalue $\mu_{\max}$. Linearizing the governing equations around the origin right after the principal eigenvalue crosses zero (e.g. $\lambda_{\max} \ll 1$) therefore gives

$$\frac{d\mathbf{x}}{dt} = \mathcal{U}|_{\mathbf{x}=\mathbf{0}} \mathbf{x}, \quad (\text{S2.84})$$

and the system will initially evolve along the direction of $\mathbf{v}_{\max}$, such that

$$\mathbf{x}(t) = c(t)\mathbf{v}_{\max}, \quad (\text{S2.85})$$

where $c(t) \in \mathbb{C}$ is a time-dependent complex coefficient. As the coupling matrix is real-valued and symmetric, the eigenvector $\mathbf{v}_{\max}$ can be chosen to be real-valued and with $\|\mathbf{v}_{\max}\|_2 = 1$. The real-valued amplitudes of the spins can therefore be expressed as

$$\mathbf{r}(t) = \rho(t)\mathbf{a}, \quad \rho(t) = |c(t)|, \quad a_i = |v_{\max,i}|, \quad (\text{S2.86})$$

where $\rho(t) \in \mathbb{R}_{\geq 0}$ is a global amplitude coefficient. The phases of the spins will initially be either $\theta_i = \arg(c)$ if $v_{\max,i} > 0$ or $\theta_i = \arg(c) + \pi$ if $v_{\max,i} < 0$.

The amplitude and phase equations of the sigmoid PM model was defined in section S1.5 as

$$\frac{dr_i}{dt} = -r_i + \tanh \left(\alpha r_i + \gamma r_i^{q-1} \cos(q\theta_i) + \beta \sum_j^N J_{ij} r_j \cos(\theta_j - \theta_i) \right) \quad (\text{S2.87})$$

$$\frac{d\theta_i}{dt} = -\gamma r_i^{q-2} \sin(q\theta_i) + \frac{\beta}{r_i} \sum_j^N J_{ij} r_j \sin(\theta_j - \theta_i) \quad (\text{S2.88})$$

As $\theta_i \in \{\arg(c), \arg(c) + \pi\}$ initially, the phase differences $\theta_j - \theta_i$ will be either 0 or π , such that $\sin(\theta_j - \theta_i) = 0 \forall i, j$. Fixed points of the phase equations, where $d\theta_i/dt = 0$, must therefore initially satisfy

$$0 = -\gamma r_i^{q-2} \sin(q\theta_i) \Rightarrow \sin(q\theta_i) = 0, \quad (\text{S2.89})$$

which has the solutions

$$\theta_i = \frac{m\pi}{q}, \quad m \in \mathbb{Z}. \quad (\text{S2.90})$$

As $\theta_i \in \{\arg(c), \arg(c) + \pi\}$, the phase of c must therefore be one of the q discrete values given by

$$\arg(c) = \frac{2k\pi}{q}, \quad k = 0, \dots, q-1, \quad (\text{S2.91})$$

corresponding to the q discrete phases of the planar Potts model in $[0, 2\pi]$.

The argument of the hyperbolic tangent in the amplitude equation can be expressed as

$$z_i = \alpha r_i + \gamma r_i^{q-1} \cos(q\theta_i) + \beta \sum_j^N J_{ij} r_j \cos(\theta_j - \theta_i) \quad (\text{S2.92})$$

$$= (\lambda_{\max} + 1) \rho a_i + \gamma s_i \rho^{q-1} a_i^{q-1}, \quad (\text{S2.93})$$

where $s_i = \cos(q\theta_i) \in \{+1, -1\}$, and where it is used that $\cos(\theta_j - \theta_i) = \text{sign}(v_i v_j) = \sigma_i \sigma_j$ with $\sigma_i = \text{sign}(v_i)$, such that

$$\sum_j^N J_{ij} r_j \cos(\theta_j - \theta_i) = \rho \sum_j^N J_{ij} a_j \sigma_i \sigma_j = \rho \sigma_i \sum_j^N J_{ij} (\sigma_j a_j) = \rho \sigma_i \mu_{\max} v_i = \mu_{\max} \rho a_i. \quad (\text{S2.94})$$

Note that $v_i = v_{\max, i} = \sigma_i a_i$. For $z_i \ll 1$, the hyperbolic tangent can be approximated to the third order as $\tanh(z_i) = z_i - z_i^3/3 + \mathcal{O}(z_i^5)$. Assuming $q > 2$ and $\rho \ll 1$, keeping only the leading order term of the cubed argument gives

$$z_i^3 = (\lambda_{\max} + 1)^3 \rho^3 a_i^3 + \mathcal{O}(\rho^{q+1}). \quad (\text{S2.95})$$

Therefore, the amplitude equation becomes

$$\frac{dr_i}{dt} = \frac{d\rho}{dt} a_i = -r_i + z_i - \frac{1}{3} z_i^3 + \mathcal{O}(z_i^5) \quad (\text{S2.96})$$

$$= -\rho a_i + (\lambda_{\max} + 1) \rho a_i + \gamma s_i \rho^{q-1} a_i^{q-1} - \frac{1}{3} (\lambda_{\max} + 1)^3 \rho^3 a_i^3 + \mathcal{O}(\rho^5, \rho^{q+1}) \quad (\text{S2.97})$$

$$(\text{S2.98})$$

Similarly to how we extracted the time-evolution of $c(t)$ in section S2.2.1, we now multiply by a_i and sum over all spins to extract the time-evolution of the global amplitude coefficient $\rho(t)$, given that $\sum_i a_i^2 = 1$:

$$\frac{d\rho}{dt} = \sum_i^N a_i \frac{d\rho}{dt} a_i = \sum_i^N a_i \frac{dr_i}{dt} \quad (\text{S2.99})$$

$$= \lambda_{\max} \rho + \gamma S_q \rho^{q-1} - \frac{1}{3} (\lambda_{\max} + 1)^3 C_3 \rho^3 + \mathcal{O}(\rho^5, \rho^{q+1}) \quad (\text{S2.100})$$

where

$$S_q = \sum_i^N s_i a_i^q = \sum_i^N \cos(q\theta_i) a_i^q, \quad C_3 = \sum_i^N a_i^4. \quad (\text{S2.101})$$

At a steady state, $d\rho/dt = 0$, and the equation for ρ , after dividing by $\rho > 0$, becomes

$$0 = \lambda_{\max} + \gamma S_q \rho^{q-2} - \frac{1}{3} (\lambda_{\max} + 1)^3 C_3 \rho^2 + \mathcal{O}(\rho^4, \rho^q). \quad (\text{S2.102})$$

The smallest power of ρ will dominate close to the origin where $\rho \ll 1$. Depending on the number of Potts states q , this gives the leading order terms as

$$\rho \approx \begin{cases} \frac{\lambda_{\max}}{\gamma |S_q|}, & q = 3, \\ \frac{\lambda_{\max}}{[\frac{1}{3} (\lambda_{\max} + 1)^3 C_3 - \gamma S_q]}, & q = 4, \\ \frac{3\lambda_{\max}}{(\lambda_{\max} + 1)^3 C_3} \approx \frac{3\lambda_{\max}}{C_3}, & q \geq 5, \end{cases} \quad (\text{S2.103})$$

Note that in all these cases, the amplitude ρ grows linearly with $\lambda_{\max}$ close to the origin.

In summary, when α or β is increased beyond the origin destabilization threshold, the fixed points will initially evolve from the origin according to

$$\mathbf{x} \approx \rho e^{i \arg c} \mathbf{v}_{\max}, \quad (\text{S2.104})$$

where $\rho \ll 1$ is given by Eq. S2.103,

$$\arg(c) = \frac{2k\pi}{q}, \quad k = 0, \dots, q-1, \quad (\text{S2.105})$$

$$\lambda_{\max} = \alpha - 1 + \beta \mu_{\max} \ll 1, \quad (\text{S2.106})$$

and $\mathbf{v}_{\max} \in \mathbb{R}^N$ is the principal eigenvector of the coupling matrix $\mathbf{J}$ with eigenvalue $\mu_{\max}$.

Thus, after the origin is destabilized, the system bifurcates and all spins initially evolve parallel or anti-parallel to one of q discrete phases, depending on the sign of the corresponding element in the principal eigenvector. The amplitude grows linearly with $\lambda_{\max}$ close to the origin. As $\lambda_{\max}$ is increased further from zero, the amplitude will deviate from this linear behavior as higher-order terms become significant.

S2.3 Amplitudes of stable fixed points

To determine whether the different Potts machine models enforce amplitude homogeneity, we analyze the amplitudes of stable non-origin fixed points. We focus on fixed points where all spin amplitudes are non-zero ($r_i > 0 \forall i$) and phases are discretized to the planar Potts states ($\theta_i = \theta_i^{(k)} = 2\pi k_i/q$ for some $k_i \in \{0, \dots, q-1\}$). This simplification allows us to isolate the amplitude dynamics from the phase dynamics.

For the polynomial model, the amplitude equation derived in supplementary materials S1.5 is

$$\frac{dr_i}{dt} = \alpha r_i - r_i^n + \gamma r_i^{q-1} \cos(q\theta_i) + \beta \sum_j^N J_{ij} r_j \cos(\theta_j - \theta_i). \quad (\text{S2.107})$$

This applies directly to the NEC model (with $\beta = 1$ and dynamic α_i), the q -PDC model (with $\alpha = -1$), and serves as the basis for the sigmoid model. The q -SHIL model has fixed amplitudes by definition, so amplitude homogeneity is trivially enforced.

S2.3.1 Fixed point amplitudes with discretized phases

Assuming phases are discretized to Potts states, $\cos(q\theta_i) = 1$ for all i , and we define the effective coupling matrix

$$\tilde{J}_{ij} = J_{ij} \cos(\Delta_{ij}), \quad \Delta_{ij} = \theta_j^{(k)} - \theta_i^{(k)} = \frac{2\pi}{q}(k_j - k_i), \quad (\text{S2.108})$$

which accounts for phase differences between spins. The amplitude equation becomes

$$\frac{dr_i}{dt} = \alpha r_i - r_i^n + \gamma r_i^{q-1} + \beta \sum_j^N \tilde{J}_{ij} r_j. \quad (\text{S2.109})$$

At a fixed point, $dr_i/dt = 0$, giving

$$r_i^n - \gamma r_i^{q-1} - \alpha r_i = \beta \sum_j^N \tilde{J}_{ij} r_j. \quad (\text{S2.110})$$

S2.3.2 Existence of homogeneous amplitude fixed points

A homogeneous amplitude fixed point has $r_i = r^* > 0$ for all i . Substituting into Eq. (S2.110) and dividing by $r^* > 0$ gives

$$(r^*)^{n-1} - \gamma (r^*)^{q-2} - \alpha = \beta \sum_j^N \tilde{J}_{ij} = \beta \tilde{\rho}_i, \quad (\text{S2.111})$$

where $\tilde{\rho}_i = \sum_j \tilde{J}_{ij}$ is the i -th row sum of the effective coupling matrix. For a homogeneous fixed point to exist for all spins simultaneously, either:

1. All row sums are equal: $\tilde{\rho}_i = \tilde{\rho} \forall i$, or
2. The coupling is weak: $\beta \rightarrow 0$.

For general coupling matrices and phase configurations, the row sums $\tilde{\rho}_i$ differ across spins, so homogeneous amplitude fixed points generically do not exist. This is the fundamental reason why models without additional mechanisms cannot enforce amplitude homogeneity.

S2.3.3 Stability of homogeneous amplitude fixed points

To analyze stability, we linearize the amplitude equations around a homogeneous fixed point $r_i = r^* + \delta r_i$ where $|\delta r_i| \ll r^*$. The Jacobian matrix of the amplitude equations with respect to the amplitudes, evaluated at the homogeneous fixed point, has elements

$$\mathcal{W}_{ij} = \left. \frac{\partial}{\partial r_j} \frac{dr_i}{dt} \right|_{r_k=r^*} = (\alpha - n(r^*)^{n-1} + (q-1)\gamma(r^*)^{q-2}) \delta_{ij} + \beta \tilde{J}_{ij}. \quad (\text{S2.112})$$

This can be written in matrix form as

$$\mathcal{W} = \lambda_{\text{self}} \mathbf{I} + \beta \tilde{\mathbf{J}}, \quad (\text{S2.113})$$

where

$$\lambda_{\text{self}} = \alpha - n(r^*)^{n-1} + (q-1)\gamma(r^*)^{q-2} \quad (\text{S2.114})$$

is the self-interaction term arising from the nonlinear amplitude dynamics. The eigenvalues of $\mathcal{W}$ are

$$\nu_k = \lambda_{\text{self}} + \beta \tilde{\mu}_k, \quad (\text{S2.115})$$

where $\tilde{\mu}_k$ are the eigenvalues of the effective coupling matrix $\tilde{\mathbf{J}}$.

For stability, all eigenvalues must be negative: $\nu_k < 0 \forall k$. This requires

$$\lambda_{\text{self}} + \beta \tilde{\mu}_{\text{max}} < 0, \quad (\text{S2.116})$$

where $\tilde{\mu}_{\text{max}}$ is the largest eigenvalue of $\tilde{\mathbf{J}}$. Rearranging gives the stability condition

$$n(r^*)^{n-1} - (q-1)\gamma(r^*)^{q-2} > \alpha + \beta \tilde{\mu}_{\text{max}}. \quad (\text{S2.117})$$

S2.3.4 Application to specific models

Polynomial and q -PDC models: For these models, the right-hand side of Eq. (S2.110) is spin-dependent through the row sums $\tilde{\rho}_i$, which generically vary across spins. Therefore, homogeneous amplitude fixed points do not exist for general coupling matrices, and amplitude homogeneity is not enforced. Even if a homogeneous fixed point existed (e.g., for specially structured coupling matrices), whether condition (S2.117) is satisfied depends on the specific values of r^* , γ , and the coupling spectrum, with no general guarantee of stability.

q -SHIL model: In this model, all amplitudes are fixed to unity by definition ($r_i = 1 \forall i$). Amplitude homogeneity is therefore trivially enforced without requiring any stability analysis.

NEC model: The NEC model augments the amplitude dynamics with spin-dependent gains α_i that evolve according to

$$\frac{d\alpha_i}{dt} = \varepsilon_\alpha (\sqrt{r_{\text{target}}} - \sqrt{r_i}), \quad (\text{S2.118})$$

where $\varepsilon_\alpha > 0$ is the gain relaxation rate and $r_{\text{target}} > 0$ is the target amplitude. At steady state, $d\alpha_i/dt = 0$ requires $r_i = r_{\text{target}}$ for all i , so the only steady state has homogeneous amplitudes.

We now linearize the coupled system around $(r_i, \alpha_i) = (r_{\text{target}}, \alpha^*)$ where α^* satisfies the amplitude fixed point equation. Writing $r_i = r_{\text{target}} + \delta r_i$ and $\alpha_i = \alpha^* + \delta \alpha_i$, the linearized dynamics become

$$\frac{d\delta r_i}{dt} = \lambda_{\text{self}} \delta r_i + r_{\text{target}} \delta \alpha_i + \sum_j \tilde{J}_{ij} \delta r_j, \quad (\text{S2.119})$$

$$\frac{d\delta \alpha_i}{dt} = -\frac{\varepsilon_\alpha}{2\sqrt{r_{\text{target}}}} \delta r_i, \quad (\text{S2.120})$$

where we used $\beta = 1$ for the NEC model. The gain dynamics introduce negative feedback: if $\delta r_i > 0$, then $d\delta \alpha_i/dt < 0$, which causes α_i to decrease. A decreasing α_i reduces dr_i/dt (since the amplitude equation has a positive $\alpha_i r_i$ term), eventually making it negative and driving r_i back toward r_{target} . The converse holds when $\delta r_i < 0$. This feedback drives all amplitudes toward r_{target} regardless of the coupling structure

Sigmoid model: The sigmoid model replaces the polynomial amplitude equation with

$$\frac{dr_i}{dt} = -r_i + \tanh \left(\alpha r_i + \gamma r_i^{q-1} \cos(q\theta_i) + \beta \sum_j^N J_{ij} r_j \cos(\theta_j - \theta_i) \right). \quad (\text{S2.121})$$

At steady state, $r_i = \tanh(R_i)$ where R_i is the argument. As the parameters α and β increase during annealing, $|R_i| \rightarrow \infty$ for all spins. When $\tanh(R_i) \rightarrow 1$, we have $dr_i/dt = -r_i + \tanh(R_i) \approx 1 - r_i$, which is positive for $r_i < 1$ and negative for $r_i > 1$, such that the amplitudes must converge to $r_i = 1$. Thus all amplitudes converge to unity regardless of coupling structure.

S2.4 Phases of stable fixed points

To determine whether the different Potts machine models enforce phase discretization, we analyze the phases of stable non-origin fixed points. We focus on fixed points where all spin amplitudes are non-zero ($r_i > 0 \forall i$), ensuring that phases are well-defined. This does not exclude the existence of other fixed points where some spins may have zero amplitude. The stable fixed-point phases are determined using the phase equations derived by expressing the complex governing equations in polar form $x_i = r_i e^{i\theta_i}$.

For the polynomial model, the derivation of the phase equations is outlined in supplementary materials S1.5. This derivation applies directly to the q -PDC model and to the NEC model when fixing β to 1. The sigmoid model uses the same phase equations per definition. The q -SHIL model phase equations (the Kuramoto model) are per definition the same as for the polynomial model with $\beta = 1$ and all amplitudes fixed to unity. So the phase equations derived for the polynomial model,

$$\frac{d\theta_i}{dt} = -\gamma r_i^{q-2} \sin(q\theta_i) + \frac{\beta}{r_i} \sum_j^N J_{ij} r_j \sin(\theta_j - \theta_i), \quad (\text{S2.122})$$

apply to all Potts machine models discussed in this work. For the NEC model, $\beta = 1$. For the q -SHIL model, $\beta = 1$ and all amplitudes are fixed to unity ($r_i = 1 \forall i$). Fixed points of the phase equations satisfy $d\theta_i/dt = 0$.

S2.4.1 Stable fixed point phases without coupling ($\beta = 0$)

When there is no coupling between spins ($\beta = 0$), the fixed point equation reduces to

$$\gamma r_i^{q-1} \sin(q\theta_i) = 0, \quad (\text{S2.123})$$

which assuming $\gamma \neq 0$ and $r_i \neq 0$ has the solutions

$$\theta_i = \frac{k\pi}{q}, \quad k \in \mathbb{Z}. \quad (\text{S2.124})$$

Stability requires

$$\frac{\partial}{\partial \theta_i} \frac{d\theta_i}{dt} = -\gamma q r_i^{q-2} \cos(q\theta_i) < 0, \quad (\text{S2.125})$$

which in $\theta_i \in [0, 2\pi]$ is fulfilled for

$$\theta_i^{(k)} = \frac{2\pi k}{q}, \quad k = 0, \dots, q-1. \quad (\text{S2.126})$$

This corresponds to the q discrete phases of the planar Potts model.

S2.4.2 Stable fixed point phases with coupling ($\beta > 0$) assuming small offset from Potts model phases

We consider the case with coupling between spins ($\beta > 0$) and analyze the stability of fixed points close to the planar Potts model phases, such that each phase can be written as a small deviation from its nearest planar Potts model phase:

$$\theta_i = \theta_i^{(k)} + \varepsilon_i, \quad |\varepsilon_i| \ll 1, \quad (\text{S2.127})$$

and let

$$\Delta_{ij} = \theta_j^{(k)} - \theta_i^{(k)} = \frac{2\pi}{q}(k_j - k_i). \quad (\text{S2.128})$$

Expanding the fixed point equation to first order in ε_i gives

$$0 = -\gamma r_i^{q-2} q \varepsilon_i + \frac{\beta}{r_i} \sum_j^N J_{ij} r_j (\sin(\Delta_{ij}) + \cos(\Delta_{ij})(\varepsilon_j - \varepsilon_i)). \quad (\text{S2.129})$$

Collecting this into a linear system for the offsets vector $\boldsymbol{\varepsilon} = (\varepsilon_1, \dots, \varepsilon_N)^T \in \mathbb{R}^N$ gives

$$(\gamma q \mathbf{D} + \beta \mathbf{L}_c) \boldsymbol{\varepsilon} = \beta \mathbf{b}, \quad (\text{S2.130})$$

where

$$\mathbf{D} = \text{diag}(r_1^{q-2}, \dots, r_N^{q-2}) \in \mathbb{R}^{N \times N}, \quad (\text{S2.131})$$

$$\mathbf{L}_c = \{(\mathbf{L}_c \mathbf{v})_i = \frac{1}{r_i} \sum_{j=1}^N J_{ij} r_j \cos(\Delta_{ij})(v_i - v_j)\} \in \mathbb{R}^{N \times N}, \quad (\text{S2.132})$$

$$\mathbf{b} = (b_1, \dots, b_N)^T \in \mathbb{R}^N, \quad b_i = \frac{1}{r_i} \sum_{j=1}^N J_{ij} r_j \sin(\Delta_{ij}). \quad (\text{S2.133})$$

The stability of a fixed point in phase space is determined by the eigenvalues of the Jacobian matrix of the phase equations, with elements

$$\mathcal{V}_{ij} = \frac{\partial}{\partial \theta_j} \frac{d\theta_i}{dt} = -\gamma q r_i^{q-2} \cos(q\theta_i) \delta_{ij} + \beta \left(\frac{1}{r_i} J_{ij} r_j \cos(\theta_j - \theta_i) - \frac{1}{r_i} \delta_{ij} \sum_k^N J_{ik} r_k \cos(\theta_k - \theta_i) \right). \quad (\text{S2.134})$$

Assuming $\gamma > 0$, the first term is negative around the planar Potts phases as $\cos(q\theta_i^{(k)}) = 1$. The second term can be positive or negative depending on the coupling matrix and the phases.

Increasing β and γ simultaneously such that their ratio remains constant, the offsets will remain constant and the stability will remain unaffected, as both the linear system in Eq. (S2.130) and the Jacobian matrix in Eq. (S2.134) are simply scaled by the same positive factor.

As $\beta \rightarrow 0$, fixed points approach the planar Potts model phases, since the linear system in Eq. (S2.130) reduces to $\gamma q \mathbf{D} \boldsymbol{\varepsilon} = 0$, which has the unique solution $\boldsymbol{\varepsilon} = \mathbf{0}$ because $\mathbf{D}$ is positive definite (we consider only fixed points with non-zero amplitudes). In this limit, the Jacobian matrix in Eq. (S2.134) becomes diagonal with negative diagonal elements, confirming that these fixed points are stable.

For $\beta > 0$, $\beta/\gamma \ll 1$, the linear system in Eq. (S2.130) can be approximated as

$$\boldsymbol{\varepsilon} \approx \frac{\beta}{\gamma q} \mathbf{D}^{-1} \mathbf{b}, \quad (\text{S2.135})$$

showing that the offsets grow linearly with the β/γ ratio when this ratio is small. Note that $\mathbf{D}$ is positive definite and therefore invertible. From Eq. (S2.134), we see that increasing β/γ will increase the magnitude of the off-diagonal coupling terms relative to the diagonal self-interaction term, which may destabilize the fixed point if the off-diagonal part has nonnegative eigenvalues.

S2.4.3 Conditions for phase discretization enforcement

The above analysis shows that stable fixed points close to the planar Potts model phases exist when the β/γ ratio is sufficiently small. An annealing scheme that gradually reduces the β/γ ratio during the optimization process will therefore enforce phase discretization, as fixed points will approach the planar Potts model phases and become stable if not already stable.

S3 Checks of dynamical properties

The three dynamical properties introduced in the main text can be evaluated directly from the governing equations assuming the annealing schedules of Table 1. Below we outline how each property is checked and then evaluate each model based on the stability and fixed-point analysis of section S2.

S3.1 How dynamical properties are checked

Connected origin-to-solution branch For models with amplitude dynamics, this property requires that the trivial fixed point at the origin is initially stable and bifurcates into a nontrivial solution branch that evolves continuously from the origin under the applied annealing schedule. As shown in section S2.1, the origin stability depends on the principal eigenvalue of the Jacobian matrix, which crosses zero at a threshold determined by the annealing parameters. Section S2.2 demonstrates how fixed points evolve after the origin loses stability. Models without amplitude dynamics do not have an origin fixed point and thus this requirement is not applicable.

Enforced amplitude homogeneity Enforcing amplitude homogeneity requires a mechanism that drives all spin amplitudes to the same value during evolution. As analyzed in section S2.3, for general coupling matrices, homogeneous amplitude fixed points generically do not exist unless additional mechanisms are present. This can be achieved through dynamic gain control (NEC model), saturating nonlinearities (sigmoid models), or by fixing the amplitudes at all times (q -SHIL model).

Enforced phase discretization Phase discretization means that, through evolution of the system, the continuous spin phases converge to the discrete set of phases corresponding to the Potts model states. As shown in section S2.4.2, stable fixed points approach the planar Potts phases $\theta_i^{(k)} = 2\pi k/q$ when the β/γ ratio is small. Decreasing this ratio during annealing ensures that fixed points both approach the Potts phases and become (or remain) stable.

S3.2 NEC model

Connected origin-to-solution branch (Fulfilled): As shown in section S2.1.1, the origin is stable for $\alpha < \alpha_{\text{th}} = -\mu_{\text{max}}$, where μ_{max} is the largest eigenvalue of the coupling matrix. The NEC gain equation (4b) increases α_i over time while $r_i < r_{\text{target}}$. As α passes α_{th} , the origin loses stability and a continuous branch of growing amplitudes emerges along one of q equispaced discrete phases, as demonstrated in section S2.2.1. The NEC model thus has a connected origin-to-solution branch.

Enforced amplitude homogeneity (Fulfilled): As analyzed in section S2.3, the NEC model employs dynamic gain control through Eq. (4b), where the spin-dependent gains α_i evolve to drive all amplitudes toward the target value r_{target} . The negative feedback mechanism ensures that at steady state, $r_i = r_{\text{target}}$ for all i , regardless of the coupling structure. Therefore, amplitude homogeneity is enforced.

Enforced phase discretization (Fulfilled): For the NEC model, $\beta = 1$ by definition. As γ is increased during annealing, the β/γ ratio decreases. By the analysis of section S2.4.2, this ensures that fixed points approach the Potts phases and remain stable. Therefore, phase discretization is enforced.

S3.3 q -PDC model

Connected origin-to-solution branch (Not fulfilled): As shown in section S2.1.1, with $\alpha = -1$ fixed, the origin is stable for $\beta < \beta_{\text{th}} = 1/\mu_{\text{max}}$. The annealing scheme fixes β while increasing γ . Since β does not cross the threshold during annealing, the origin never loses stability and no connected origin-to-solution branch exists.

Enforced amplitude homogeneity (Not fulfilled): As shown in section S2.3, for general coupling matrices, the amplitude fixed point equation (S2.110) requires spin-dependent row sums of the effective coupling matrix to be equal for a homogeneous fixed point to exist. Since this is generically not satisfied, the q -PDC model does not enforce amplitude homogeneity.

Enforced phase discretization (Fulfilled): With β fixed and γ increasing during annealing, the β/γ ratio decreases. By section S2.4.2, phase discretization is therefore enforced.

S3.4 q -SHIL model

Connected origin-to-solution branch (Not applicable): The q -SHIL model does not have amplitude dynamics (all amplitudes are fixed to unity by definition) and thus does not have an origin fixed point. Therefore, the requirement of a connected origin-to-solution branch is not applicable.

Enforced amplitude homogeneity (Fulfilled): As noted in section S2.3, the q -SHIL model has all amplitudes fixed to unity ($r_i = 1 \forall i$) by definition. Therefore, amplitude homogeneity is trivially enforced.

Enforced phase discretization (Fulfilled): For the q -SHIL model, $\beta = 1$ and $r_i = 1 \forall i$ by definition. As γ is increased during annealing, the β/γ ratio decreases. By section S2.4.2, phase discretization is therefore enforced.

S3.5 Polynomial PM model

Connected origin-to-solution branch (Fulfilled): As shown in section S2.1.1, with $\alpha = -1$ fixed, the origin is stable for $\beta < \beta_{\text{th}} = 1/\mu_{\text{max}}$. As β is increased past β_{th} during annealing, the origin loses stability and a continuous branch of growing amplitudes emerges along one of q equispaced discrete phases, as demonstrated in section S2.2.1. The polynomial PM model thus has a connected origin-to-solution branch.

Enforced amplitude homogeneity (Not fulfilled): As shown in section S2.3, for general coupling matrices, homogeneous amplitude fixed points generically do not exist because the row sums of the effective coupling matrix differ across spins. The polynomial PM model does not include any mechanism to enforce amplitude homogeneity.

Enforced phase discretization (Not fulfilled): Both β and γ can be controlled during annealing. Phase discretization can be enforced by increasing γ faster than linearly with respect to β , e.g., $\gamma = \beta^p$ with $p > 1$, which decreases the β/γ ratio. However, with the employed annealing scheme where γ and β increase linearly together ($\gamma \propto \beta$), the β/γ ratio remains constant. By the analysis of section S2.4.2, phase discretization is therefore not enforced with this schedule.

Still, as illustrated in figure 1 in the main text, the phases still cluster around the Potts phases for the employed annealing scheme, if not discretizing completely.

S3.6 Sigmoid PM model

Connected origin-to-solution branch (Fulfilled): As shown in section S2.1.2, with $\alpha = 1$ fixed, the origin is stable for $\beta < \beta_{\text{th}} = 1/\mu_{\text{max}}$ (from Eq. (S2.65) with $\alpha = 1$, giving $\beta_{\text{th}} = (1 - \alpha)/\mu_{\text{max}} = 0$; however, for $\alpha > 1$ during annealing, the threshold becomes positive). As β is increased past β_{th} during annealing, the origin loses stability and a continuous branch of growing amplitudes emerges along one of q equispaced discrete phases, as demonstrated in section S2.2.2. The sigmoid PM model thus has a connected origin-to-solution branch.

Enforced amplitude homogeneity (Fulfilled): As analyzed in section S2.3, the sigmoid model uses the saturating nonlinearity $\tanh(\cdot)$, which bounds outputs to $(-1, 1)$. As the gain parameters α and β increase during annealing, the argument of the hyperbolic tangent grows large, driving $\tanh(R_i) \rightarrow 1$ for all spins. Since steady states satisfy $r_i = \tanh(R_i)$, all amplitudes converge to unity regardless of coupling structure. Therefore, amplitude homogeneity is enforced.

Enforced phase discretization (Not fulfilled): With the employed annealing scheme where γ and β increase linearly together, the β/γ ratio remains constant. By section S2.4.2, phase discretization is therefore not enforced with this schedule. However, the phases still cluster around the Potts phases, as illustrated in figure 1 in the main text.

S4 Requirements on nonlinearity order to ensure amplitude bounding in polynomial-based Potts machines

For three of the five studied Potts machine (PM) models, namely the NEC, q -PDC, and polynomial PM models, the governing equation for the real-valued amplitude r of an uncoupled spin can be written on the following form (see equation S2.107):

$$\frac{dr}{dt} = \alpha r + -r^n + \gamma r^{q-1} \cos q\theta \quad (\text{S4.136})$$

Without coupling, the spin phase will quickly converge to a discrete Potts state, such that $\cos q\theta = 1$ as shown in section S2.4.2. Thus, the values of n and q determine the bounding of the amplitude r in the following way:

$$\begin{array}{llll} n > q - 1 & \rightarrow & r^n \text{ grows faster than } \gamma r^{q-1} & \rightarrow \text{Amplitude always bounded} \\ n = q - 1 & \rightarrow & \gamma r^{q-1} \text{ grows faster than } r^n \text{ for } \gamma > 1 & \rightarrow \text{Amplitude runaway for } \gamma > 1 \\ n < q - 1 & \rightarrow & \gamma r^{q-1} \text{ grows faster than } r^n \text{ for } \gamma > 0 & \rightarrow \text{Amplitude runaway for } \gamma > 0 \end{array}$$

To ensure that amplitudes are bounded for any value of γ , we must therefore require $n \geq q$. It should be noted, however, that this analysis does not consider coupling, which can assist in bounding the amplitudes.

S5 Time-domain simulations with the Euler-Maruyama method

The Potts and Ising machine models can be expressed as multi-dimensional stochastic differential equations (SDEs) of the form:

$$\frac{d\mathbf{x}}{dt} = \mathbf{F}(\mathbf{x}, t) + f_{\xi} \xi(t) \quad (\text{S5.137})$$

where $\mathbf{x}$ is the state vector representing the spin amplitudes and/or phases, $\mathbf{F}(\mathbf{x}, t)$ is the drift term representing the deterministic dynamics, $\xi(t)$ represents Gaussian white noise, and f_{ξ} is a noise factor scaling the intensity.

Given a time step size Δt , the Euler-Maruyama method [28] discretizes the integration of the equation as:

$$\mathbf{x}[k+1] = \mathbf{x}[k] + \mathbf{F}(\mathbf{x}[k], k)\Delta t + f_{\xi}\Delta\xi[k] \quad (\text{S5.138})$$

where the stochastic process increment $\Delta\xi[k]$ is sampled from a normal distribution with mean zero and variance Δt . For models with complex-valued spins, the stochastic process is applied independently to both the real and imaginary parts, with variance $\Delta t/2$ for each part, such that the total variance is still Δt .

S6 Specification of graph for qualitative dynamics analysis

Listing 1: Specification of the graph used for the qualitative dynamics analysis shown in Figure 1. The graph has 20 vertices with 50% edge probability, generated using the Rudy graph generator [31]. The format is the DIMACS graph format, with comments (c), problem definition (p), and edges (e).

```

1 c Optimum cut value (max-3-cut): 84
2 c Optimum energy (max-3-cut): 12
3 c
4 c Average absolute coupling: 1.0
5 c Maximum eigenvalue: 4.067997860365148
6 p 20 96 1
7 e 1 2 1
8 e 1 4 1
9 e 1 6 1
10 e 1 7 1
11 e 1 9 1
12 e 1 13 1
13 e 1 14 1
14 e 1 16 1
15 e 1 17 1
16 e 1 20 1
17 e 2 3 1
18 e 2 4 1
19 e 2 9 1
20 e 2 11 1
21 e 2 12 1
22 e 2 14 1
23 e 2 15 1
24 e 2 17 1
25 e 2 19 1
26 e 2 20 1
27 e 3 4 1
28 e 3 7 1
29 e 3 10 1
30 e 3 11 1
31 e 3 12 1
32 e 3 15 1
33 e 3 16 1
34 e 3 20 1
35 e 4 5 1
36 e 4 7 1
37 e 4 8 1
38 e 4 11 1
39 e 4 13 1
40 e 4 14 1
41 e 4 18 1
42 e 5 6 1
43 e 5 7 1
44 e 5 9 1
45 e 5 15 1
46 e 5 17 1
47 e 5 18 1
48 e 5 19 1
49 e 5 20 1
50 e 6 7 1
51 e 6 9 1
52 e 6 10 1
53 e 6 12 1
54 e 6 13 1
55 e 6 14 1
56 e 6 16 1
57 e 6 17 1
58 e 6 20 1
59 e 7 8 1

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60 e 7 12 1
61 e 7 14 1
62 e 7 16 1
63 e 7 18 1
64 e 7 20 1
65 e 8 9 1
66 e 8 12 1
67 e 8 18 1
68 e 8 20 1
69 e 9 10 1
70 e 9 13 1
71 e 9 15 1
72 e 9 16 1
73 e 9 17 1
74 e 9 19 1
75 e 10 13 1
76 e 10 14 1
77 e 10 15 1
78 e 10 16 1
79 e 10 18 1
80 e 10 19 1
81 e 11 14 1
82 e 11 15 1
83 e 11 17 1
84 e 11 18 1
85 e 11 19 1
86 e 12 15 1
87 e 12 16 1
88 e 13 14 1
89 e 13 15 1
90 e 13 17 1
91 e 13 18 1
92 e 13 19 1
93 e 14 15 1
94 e 14 17 1
95 e 14 19 1
96 e 14 20 1
97 e 15 17 1
98 e 16 17 1
99 e 16 18 1
100 e 17 19 1
101 e 17 20 1
102 e 18 20 1

```

S7 Simulation and optimization parameters

In the following section, parameter sweeps are denoted by $start : step : end$, where $start$ is the first value, $step$ is the increment between values, and end is the last value. For example, $1 : 0.1 : 2$ means that sweep is equispaced between 1 and 2 in increments of 0.1, i.e. $1, 1.1, 1.2, \dots, 2$.

S7.1 Non-Equilibrium Condensates (NEC) model

Table S7.1: NEC model default parameters. These parameters are used if nothing else is specified.

Symbol	Meaning	Value
T	Total simulation time	5×10^3
Δt	Time step size	10^{-2}
n_{steps}	Number of time steps	5×10^5
n_{runs}	Number of runs per hyperparameter combination	100
f_{ξ}	Noise factor	10^{-4}
ε_{α}	Gain relaxation rate	10^{-2}
α_0	Initial value of α	α_{th}
α_{th}	Threshold value of α for non-zero fixed points	Eq. (S2.39)
γ	Strength of phase discretization	$\gamma_0 + \varepsilon_{\gamma} t$
γ_0	Initial value of γ	0

Table S7.2: NEC model parameters for dynamic property study, i.e. figure 1 in the main text.

Symbol	Meaning	Value
T	Total simulation time	10^3
n_{steps}	Number of time steps	10^5
n	Nonlinearity order	3
ε_{α}	Gain relaxation rate	4×10^{-2}
ε_{γ}	Rate of increase of γ	3×10^{-3}
r_{target}	Target amplitude	1

Table S7.3: NEC model Max-3-Cut benchmark optimization sweep parameters.

Symbol	Meaning	Value
n	Nonlinearity order	3
r_{target}	Target amplitude	1.5 : 0.1 : 2.5
ε_{γ}	Rate of increase of γ	$2 \times 10^{-4} : 1 \times 10^{-4} : 3 \times 10^{-3}$

Table S7.4: NEC model Max-4-Cut benchmark optimization sweep parameters.

Symbol	Meaning	Value
n	Nonlinearity order	4
r_{target}	Target amplitude	1 : 0.2 : 3
ε_{γ}	Rate of increase of γ	$0 : 4 \times 10^{-5} : 4 \times 10^{-4}$

S7.2 q -Photon Down-Conversion (q -PDC) model

Table S7.5: q -PDC model default parameters. These parameters are used if nothing else is specified.

Symbol	Meaning	Value
T	Total simulation time	50
Δt	Time step size	10^{-3}
n_{steps}	Number of time steps	5×10^4
n_{runs}	Number of runs per hyperparameter combination	100
f_{ξ}	Noise factor	10^{-4}
β	Global coupling strength	Eq. (5b)
κ_l	Intrinsic loss	1 as per [13]
γ	Strength of phase discretization	$\gamma_0 + \varepsilon_{\gamma} t$
γ_0	Initial value of γ	0
γ_{th}	Threshold value of γ for non-zero fixed points	$\frac{4}{3^{3/4}}$

Table S7.6: q -PDC model parameters for dynamic property study, i.e. figure 1 in the main text.

Symbol	Meaning	Value
T	Total simulation time	10
n_{steps}	Number of time steps	10^4
ε_{γ}	Rate of increase of γ	$0.5\gamma_{\text{th}}$

Table S7.7: q -PDC model Max-3-Cut and Max-4-Cut benchmark optimization sweep parameters.

Symbol	Meaning	Value
ε_{γ}	Rate of increase of γ	$0 : 0.04\gamma_{\text{th}} : \gamma_{\text{th}}$

S7.3 q^{th} -harmonic Sub-Harmonic Injection Locking (q -SHIL) model

Table S7.8: q -SHIL model default parameters. These parameters are used if nothing else is specified.

Symbol	Meaning	Value
T	Total simulation time	10^3
Δt	Time step size	10^{-3}
n_{steps}	Number of time steps	10^6
n_{runs}	Number of runs per hyperparameter combination	100
f_{ξ}	Noise factor	10^{-4}
γ	Strength of phase discretization	$\gamma_0 + \varepsilon_{\gamma} t$
γ_0	Initial value of γ	0
ε_{γ}	Rate of increase of γ	10^{-2}
θ_0	Initial phase vector	Chosen from uniform distribution over $[-\pi, \pi]$

Table S7.9: q -SHIL model parameters for dynamic property study, i.e. figure 1 in the main text.

Symbol	Meaning	Value
Δt	Time step size	10^{-2}
n_{steps}	Number of time steps	10^5
ε_γ	Rate of increase of γ	2×10^{-3}

S7.4 Polynomial Potts machine model

Table S7.10: Polynomial Potts machine model default parameters. These parameters are used if nothing else is specified.

Symbol	Meaning	Value
T	Total simulation time	10^3
Δt	Time step size	10^{-3}
n_{steps}	Number of time steps	10^6
n_{runs}	Number of runs per hyperparameter combination	100
f_ξ	Noise factor	10^{-4}
β	Global coupling strength	$\beta_0 + \varepsilon_\beta t$
β_0	Initial value of β	β_{th}
β_{th}	Threshold value of β for non-zero fixed points	Eq. (S2.39)
ε_β	Rate of increase of β	2×10^{-3}
γ	Strength of phase discretization	f_γ/β

Table S7.11: Polynomial Potts machine model parameters for dynamic property study, i.e. figure 1 in the main text.

Symbol	Meaning	Value
Δt	Time step size	10^{-2}
n_{steps}	Number of time steps	10^5
n	Nonlinearity order	3
ε_β	Rate of increase of β	2×10^{-4}
f_γ	Phase discretization vs. coupling factor	1

Table S7.12: Polynomial Potts machine model Max-3-Cut benchmark optimization sweep parameters.

Symbol	Meaning	Value
n	Nonlinearity order	3 : 1 : 11
f_γ	Phase discretization vs. coupling factor	0 : 0.2 : 3

Table S7.13: Polynomial Potts machine model Max-4-Cut benchmark optimization sweep parameters.

Symbol	Meaning	Value
n	Nonlinearity order	4 : 1 : 11
f_γ	Phase discretization versus coupling factor	0 : 5×10^{-2} : 1

S7.5 Sigmoid Potts machine model

Table S7.14: Sigmoid Potts machine model default parameters. These parameters are used if nothing else is specified.

Symbol	Meaning	Value
T	Total simulation time	2×10^3
Δt	Time step size	10^{-3}
n_{steps}	Number of time steps	10^6
n_{runs}	Number of runs per hyperparameter combination	100
f_{ξ}	Noise factor	10^{-4}
β	Global coupling strength	$\beta_0 + \varepsilon_{\beta} t$
β_0	Initial value of β	β_{th}
β_{th}	Threshold value of β for non-zero fixed points	Eq. (S2.65)
ε_{β}	Rate of increase of β	10^{-2}
γ	Strength of phase discretization	$f_{\gamma}\beta$

Table S7.15: Sigmoid Potts machine model parameters for dynamic property study, i.e. figure 1 in the main text.

Symbol	Meaning	Value
T	Total simulation time	10^3
Δt	Time step size	10^{-2}
n_{steps}	Number of time steps	10^5
ε_{β}	Rate of increase of β	2×10^{-3}
f_{γ}	Phase discretization versus coupling factor	2

Table S7.16: Sigmoid Potts machine model Max-3-Cut benchmark optimization sweep parameters.

Symbol	Meaning	Value
α	Amplitude gain	$-50 : 10 : 0$
f_{γ}	Phase discretization versus coupling factor	$0 : 0.5 : 8$

The sweep of α is limited to a minimum value of -50 to limit computational time, as lower values slows spin dynamics as mentioned in the main text section 2.5.

Table S7.17: Sigmoid Potts machine model Max-4-Cut benchmark optimization sweep parameters.

Symbol	Meaning	Value
α	Amplitude gain	$-50 : 20 : 10$
f_{γ}	Phase discretization versus coupling factor	$0 : 0.1 : 2$

The sweep of α is limited to a minimum value of -50 to limit computational time, as lower values slows spin dynamics as mentioned in the main text section 2.5.

S7.6 Sigmoid Ising machine model

Table S7.18: Sigmoid Ising machine model default parameters. These parameters are used if nothing else is specified.

Symbol	Meaning	Value
T	Total simulation time	2×10^3
Δt	Time step size	10^{-3}
n_{steps}	Number of time steps	10^6
n_{runs}	Number of runs per hyperparameter combination	100
f_{ξ}	Noise factor	10^{-4}
α	Amplitude gain	-50
β	Global coupling strength	$\beta_0 + \varepsilon_{\beta} t$
β_0	Initial value of β	0
ε_{β}	Rate of increase of β	10^{-5}

The value of α is set to -50, and not lower, in order to limit computational time, as lower values slows spin dynamics as mentioned in the main text section 2.6.

Table S7.19: Sigmoid Ising machine model parameters for dynamic property study, i.e. figure 1 in the main text.

Symbol	Meaning	Value
T	Total simulation time	10^3
Δt	Time step size	10^{-2}
n_{steps}	Number of time steps	10^5
α	Amplitude gain	0.9
ε_{β}	Rate of increase of β	4×10^{-4}
B/A	Edge- vs. one-hot constraint factor (see [26])	1
ζ	External field scaling factor (see [26])	0.5

Table S7.20: Sigmoid Ising machine model Max-3-Cut benchmark optimization sweep parameters.

Symbol	Meaning	Value
B/A	Edge- vs. one-hot constraint factor (see [26])	$100n_{\text{vertices}}^{-1} : 10n_{\text{vertices}}^{-1} : 200n_{\text{vertices}}^{-1}$
ζ	External field scaling factor (see [26])	$0.4 : 0.05 : 0.7$

Table S7.21: Sigmoid Ising machine model Max-4-Cut benchmark optimization sweep parameters.

Symbol	Meaning	Value
B/A	Edge- vs. one-hot constraint factor (see [26])	$150n_{\text{vertices}} : 10n_{\text{vertices}} : 250n_{\text{vertices}}$
ζ	External field scaling factor (see [26])	$0.5 : 0.05 : 0.8$

S8 Benchmark results

S8.1 Optimality gap distribution for best hyperparameter combinations

Table S8.22: Benchmark results for studied Potts machine and Ising machine models on Max-3-Cut problems with best hyperparameter combination.

Model	Graph	Hyperparameter 1	Hyperparameter 2	Mean relative optimality gap (%)	Median relative optimality gap (%)	Minimum relative optimality gap (%)	SD on relative optimality gap (%)
NEC	G1	$f_\gamma = 8.00$	$r_{\text{target}} = 2.10$	0.25	0.24	0.16	0.05
	G2	$f_\gamma = 14.50$	$r_{\text{target}} = 1.90$	0.25	0.23	0.13	0.11
	G3	$f_\gamma = 12.50$	$r_{\text{target}} = 2.20$	0.21	0.17	0.13	0.08
	G4	$f_\gamma = 10.50$	$r_{\text{target}} = 2.20$	0.26	0.20	0.14	0.09
	G5	$f_\gamma = 10.00$	$r_{\text{target}} = 2.10$	0.26	0.25	0.15	0.05
q -PDC	G1	$\varepsilon_\gamma/\gamma_{\text{th}} = 0.48$	—	4.01	4.04	3.37	0.24
	G2	$\varepsilon_\gamma/\gamma_{\text{th}} = 0.48$	—	4.02	4.01	3.34	0.25
	G3	$\varepsilon_\gamma/\gamma_{\text{th}} = 0.48$	—	4.05	4.06	3.38	0.24
	G4	$\varepsilon_\gamma/\gamma_{\text{th}} = 0.48$	—	4.07	4.10	3.54	0.24
	G5	$\varepsilon_\gamma/\gamma_{\text{th}} = 0.48$	—	4.15	4.15	3.69	0.25
q -SHIL	G1	—	—	1.45	1.45	0.99	0.16
	G2	—	—	1.42	1.42	0.97	0.17
	G3	—	—	1.44	1.44	1.01	0.19
	G4	—	—	1.40	1.40	1.07	0.15
	G5	—	—	1.47	1.48	1.02	0.16
Polynomial PM	G1	$n = 3.00$	$f_\gamma = 2.80$	0.85	0.84	0.42	0.15
	G2	$n = 3.00$	$f_\gamma = 2.80$	0.84	0.85	0.57	0.11
	G3	$n = 3.00$	$f_\gamma = 2.80$	0.78	0.76	0.47	0.16
	G4	$n = 3.00$	$f_\gamma = 3.80$	0.85	0.86	0.58	0.12
	G5	$n = 3.00$	$f_\gamma = 3.00$	0.81	0.80	0.45	0.16
Sigmoid PM	G1	$f_\gamma = 5.00$	$\alpha = -50.00$	0.67	0.68	0.35	0.12
	G2	$f_\gamma = 5.50$	$\alpha = -50.00$	0.61	0.60	0.44	0.07
	G3	$f_\gamma = 4.50$	$\alpha = -50.00$	0.49	0.36	0.34	0.17
	G4	$f_\gamma = 5.00$	$\alpha = -40.00$	0.53	0.57	0.40	0.09
	G5	$f_\gamma = 3.50$	$\alpha = -50.00$	0.59	0.56	0.56	0.08
Sigmoid IM	G1	$(B/A)n_{\text{vertices}} = 200.00$	$\zeta = 0.55$	0.09	0.09	0.00	0.05
	G2	$(B/A)n_{\text{vertices}} = 200.00$	$\zeta = 0.55$	0.12	0.11	0.05	0.04
	G3	$(B/A)n_{\text{vertices}} = 190.00$	$\zeta = 0.55$	0.08	0.08	0.01	0.04
	G4	$(B/A)n_{\text{vertices}} = 200.00$	$\zeta = 0.50$	0.14	0.14	0.00	0.05
	G5	$(B/A)n_{\text{vertices}} = 200.00$	$\zeta = 0.50$	0.06	0.03	0.00	0.07

Table S8.23: Benchmark results for studied Potts machine and Ising machine models on Max-4-Cut problems with best hyperparameter combination.

Model	Graph	Hyperparameter 1	Hyperparameter 2	Mean relative optimality gap (%)	Median relative optimality gap (%)	Minimum relative optimality gap (%)	SD on relative optimality gap (%)
NEC	G1	$f_\gamma = 1.80$	$r_{\text{target}} = 2.60$	3.32	3.29	3.12	0.11
	G2	$f_\gamma = 0.40$	$r_{\text{target}} = 2.60$	3.32	3.28	3.28	0.12
	G3	$f_\gamma = 1.20$	$r_{\text{target}} = 2.00$	3.24	3.21	3.17	0.08
	G4	$f_\gamma = 0.60$	$r_{\text{target}} = 2.20$	3.15	3.15	3.15	0.04
	G5	$f_\gamma = 0.80$	$r_{\text{target}} = 1.80$	3.35	3.32	3.21	0.12
q -PDC	G1	$\varepsilon_\gamma/\gamma_{\text{th}} = 0.20$	—	6.03	6.00	5.35	0.29
	G2	$\varepsilon_\gamma/\gamma_{\text{th}} = 0.20$	—	6.03	6.03	5.23	0.28
	G3	$\varepsilon_\gamma/\gamma_{\text{th}} = 0.24$	—	6.06	5.99	5.33	0.36
	G4	$\varepsilon_\gamma/\gamma_{\text{th}} = 0.20$	—	6.01	5.94	5.17	0.40
	G5	$\varepsilon_\gamma/\gamma_{\text{th}} = 0.44$	—	6.08	6.05	5.29	0.35
q -SHIL	G1	—	—	3.69	3.69	3.29	0.18
	G2	—	—	3.68	3.68	3.22	0.15
	G3	—	—	3.65	3.66	3.25	0.17
	G4	—	—	3.66	3.66	3.23	0.19
	G5	—	—	3.67	3.68	3.07	0.17
Polynomial PM	G1	$n = 9.00$	$f_\gamma = 0.55$	3.66	3.66	3.22	0.18
	G2	$n = 10.00$	$f_\gamma = 0.55$	3.64	3.64	3.18	0.18
	G3	$n = 8.00$	$f_\gamma = 0.75$	3.55	3.52	3.12	0.20
	G4	$n = 10.00$	$f_\gamma = 0.20$	3.57	3.56	3.21	0.17
	G5	$n = 9.00$	$f_\gamma = 0.55$	3.63	3.62	3.16	0.19
Sigmoid PM	G1	$f_\gamma = 0.30$	$\alpha = -50.00$	3.49	3.49	3.49	0.00
	G2	$f_\gamma = 1.10$	$\alpha = -50.00$	3.43	3.43	3.43	0.00
	G3	$f_\gamma = 1.00$	$\alpha = -30.00$	3.28	3.28	3.28	0.03
	G4	$f_\gamma = 1.20$	$\alpha = -30.00$	3.32	3.32	3.32	0.00
	G5	$f_\gamma = 0.30$	$\alpha = -30.00$	3.29	3.29	3.29	0.00
Sigmoid IM	G1	$(B/A)n_{\text{vertices}} = 220.00$	$\zeta = 0.75$	0.18	0.19	0.02	0.06
	G2	$(B/A)n_{\text{vertices}} = 220.00$	$\zeta = 0.75$	0.19	0.20	0.04	0.06
	G3	$(B/A)n_{\text{vertices}} = 220.00$	$\zeta = 0.75$	0.17	0.17	0.05	0.06
	G4	$(B/A)n_{\text{vertices}} = 220.00$	$\zeta = 0.75$	0.18	0.18	0.05	0.05
	G5	$(B/A)n_{\text{vertices}} = 220.00$	$\zeta = 0.75$	0.18	0.18	0.04	0.06

S8.2 Performance sensitivity to hyperparameter settings

S8.2.1 Max-3-Cut problems

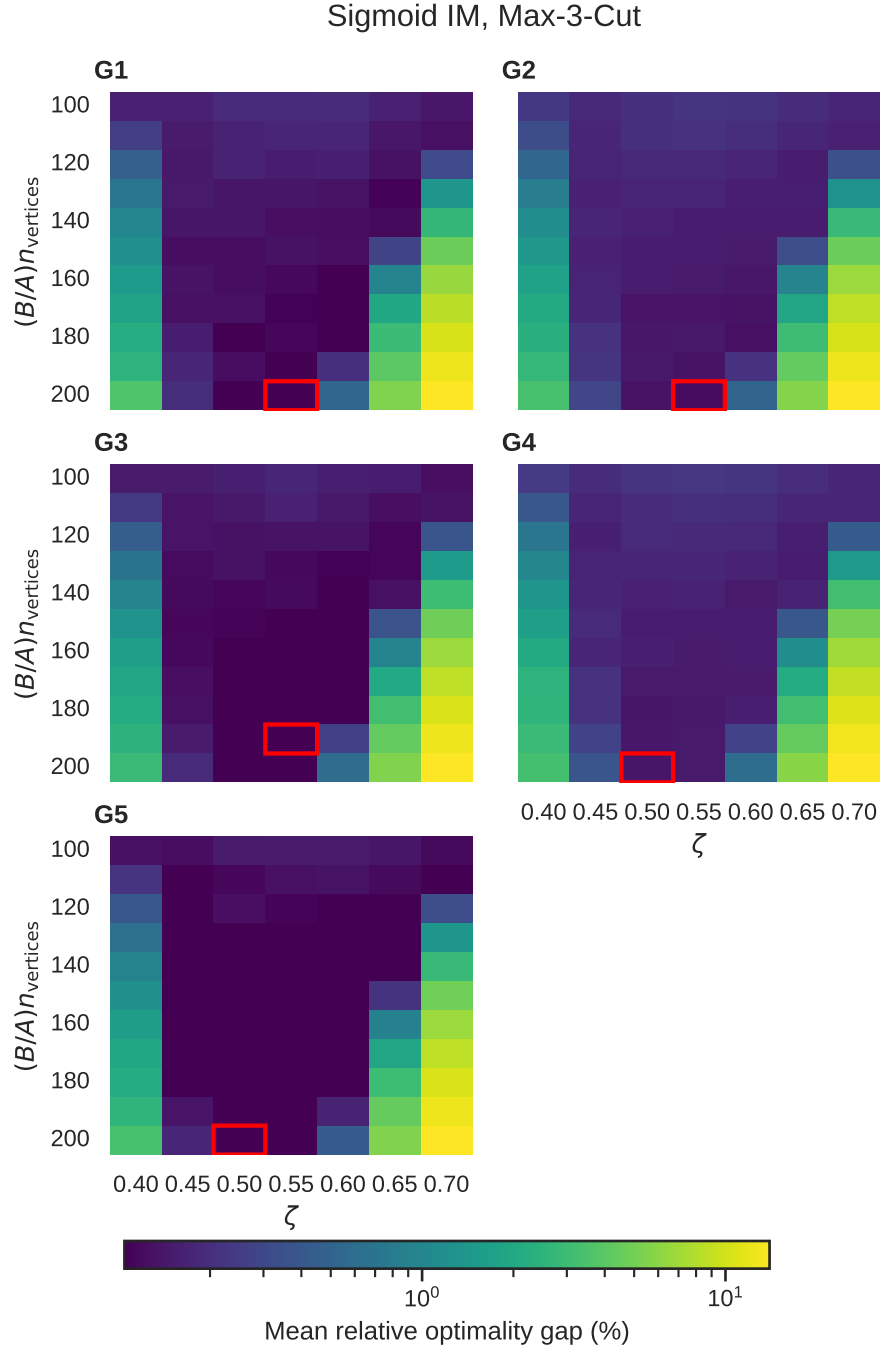


Figure S8.1: Heatmap showing the mean relative optimality gap (%) for the sigmoid IM model over the G1-G5 Max-3-Cut problems as a function of the hyperparameters α and $(B/A)n_{\text{vertices}}$. The values of the mean relative optimality gap are shown in each cell of the heatmap. The hyperparameter configuration with the minimum mean relative optimality gap is marked by a red border.

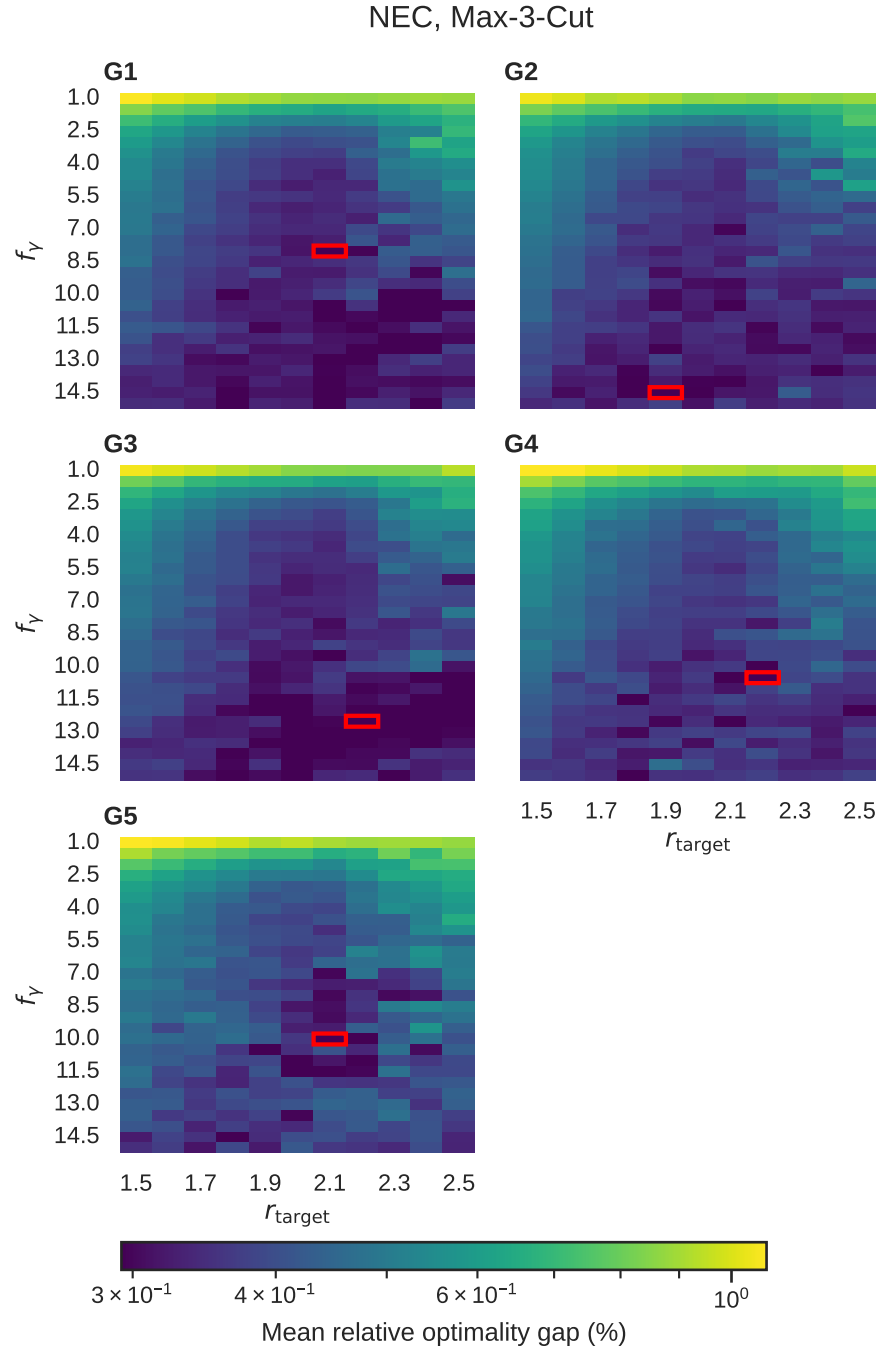


Figure S8.2: Heatmap showing the mean relative optimality gap (%) for the NEC model over the G1-G5 Max-3-Cut problems as a function of the hyperparameters f_γ and r_{target} . The values of the mean relative optimality gap are shown in each cell of the heatmap. The hyperparameter configuration with the minimum mean relative optimality gap is marked by a red border.

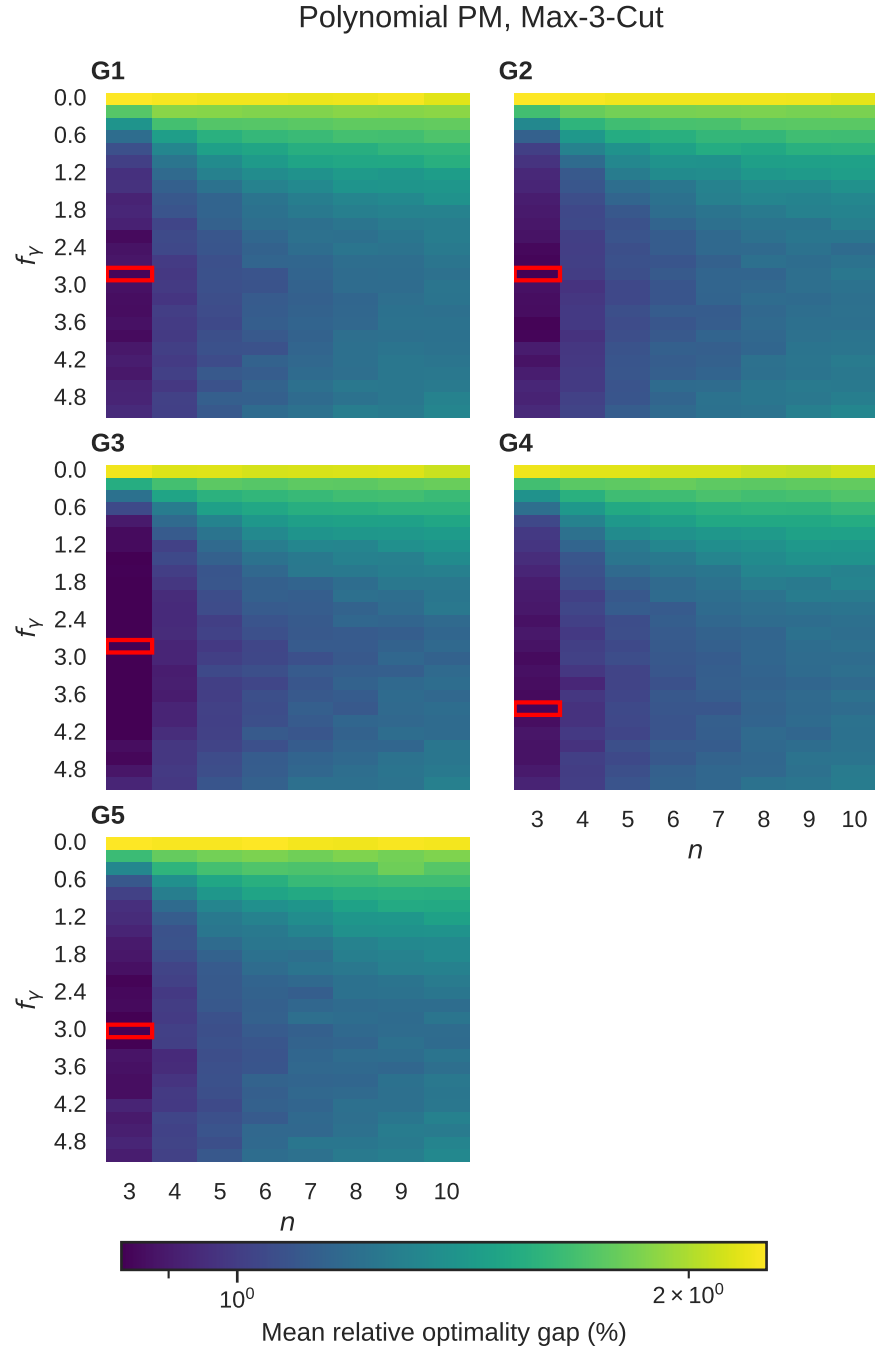


Figure S8.3: Heatmap showing the mean relative optimality gap (%) for the polynomial PM model over the G1-G5 Max-3-Cut problems as a function of the hyperparameters n and f_γ . The values of the mean relative optimality gap are shown in each cell of the heatmap. The hyperparameter configuration with the minimum mean relative optimality gap is marked by a red border.

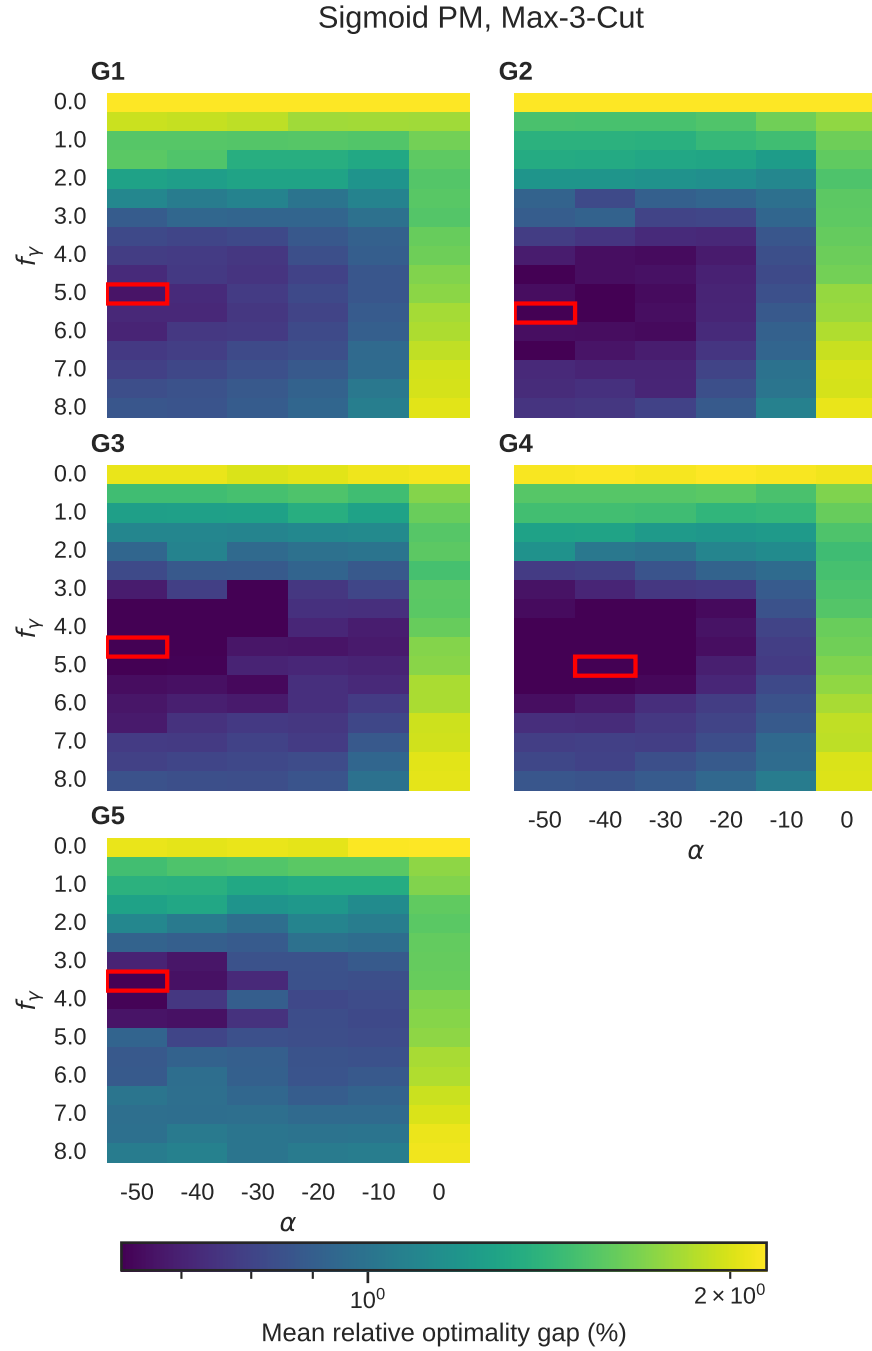


Figure S8.4: Heatmap showing the mean relative optimality gap (%) for the sigmoid PM model over the G1-G5 Max-3-Cut problems as a function of the hyperparameters f_γ and α . The values of the mean relative optimality gap are shown in each cell of the heatmap. The hyperparameter configuration with the minimum mean relative optimality gap is marked by a red border. The sweep of α is limited to a minimum value of -50 to limit computational time, as lower values slows spin dynamics as mentioned in the main text section 2.5.

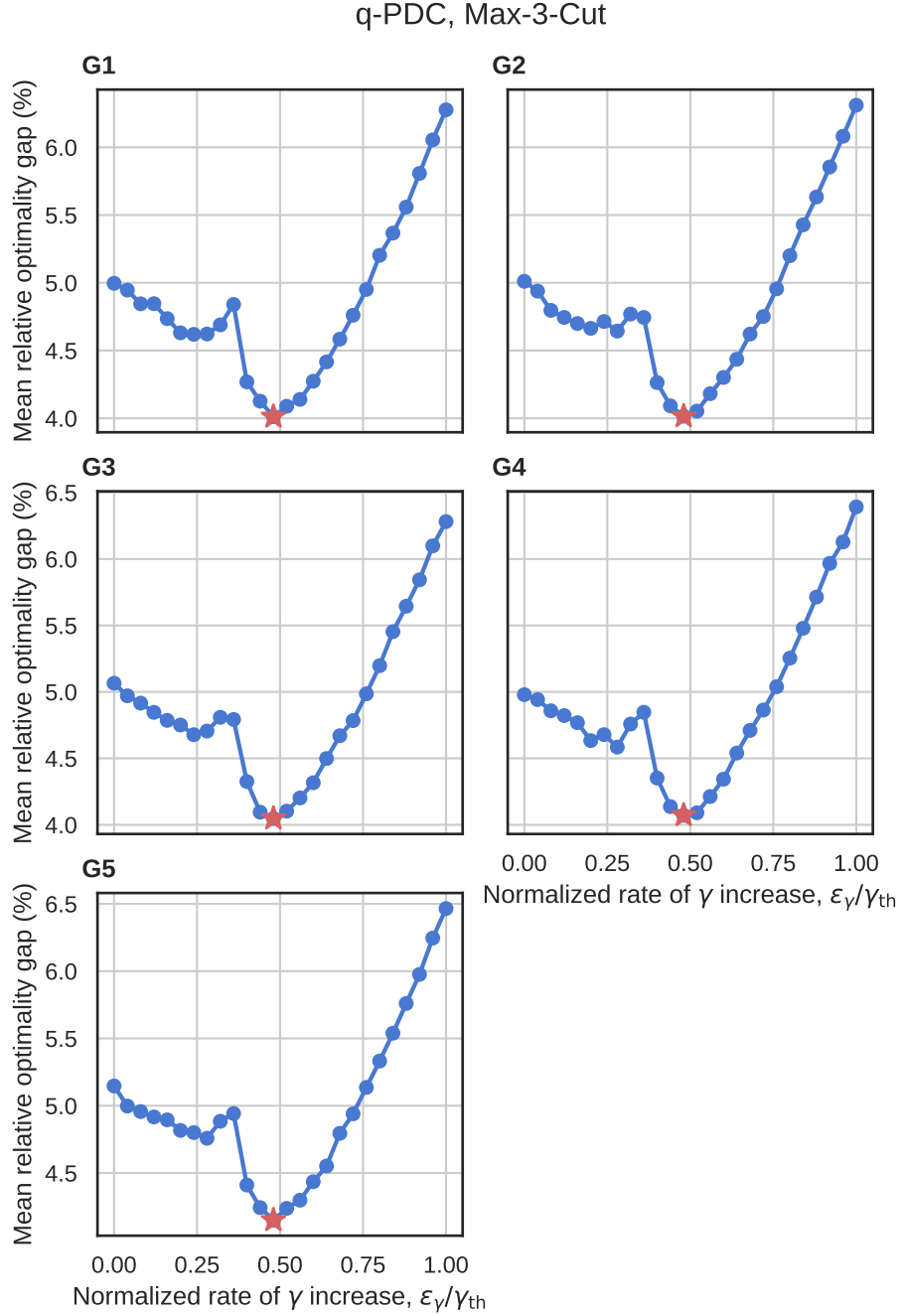


Figure S8.5: Mean relative optimality gap (%) for the q -PDC model over the G1-G5 Max-3-Cut problems as a function of the hyperparameter f_γ . The minimum mean relative optimality gap is marked by a red star.

S8.2.2 Max-4-Cut problems

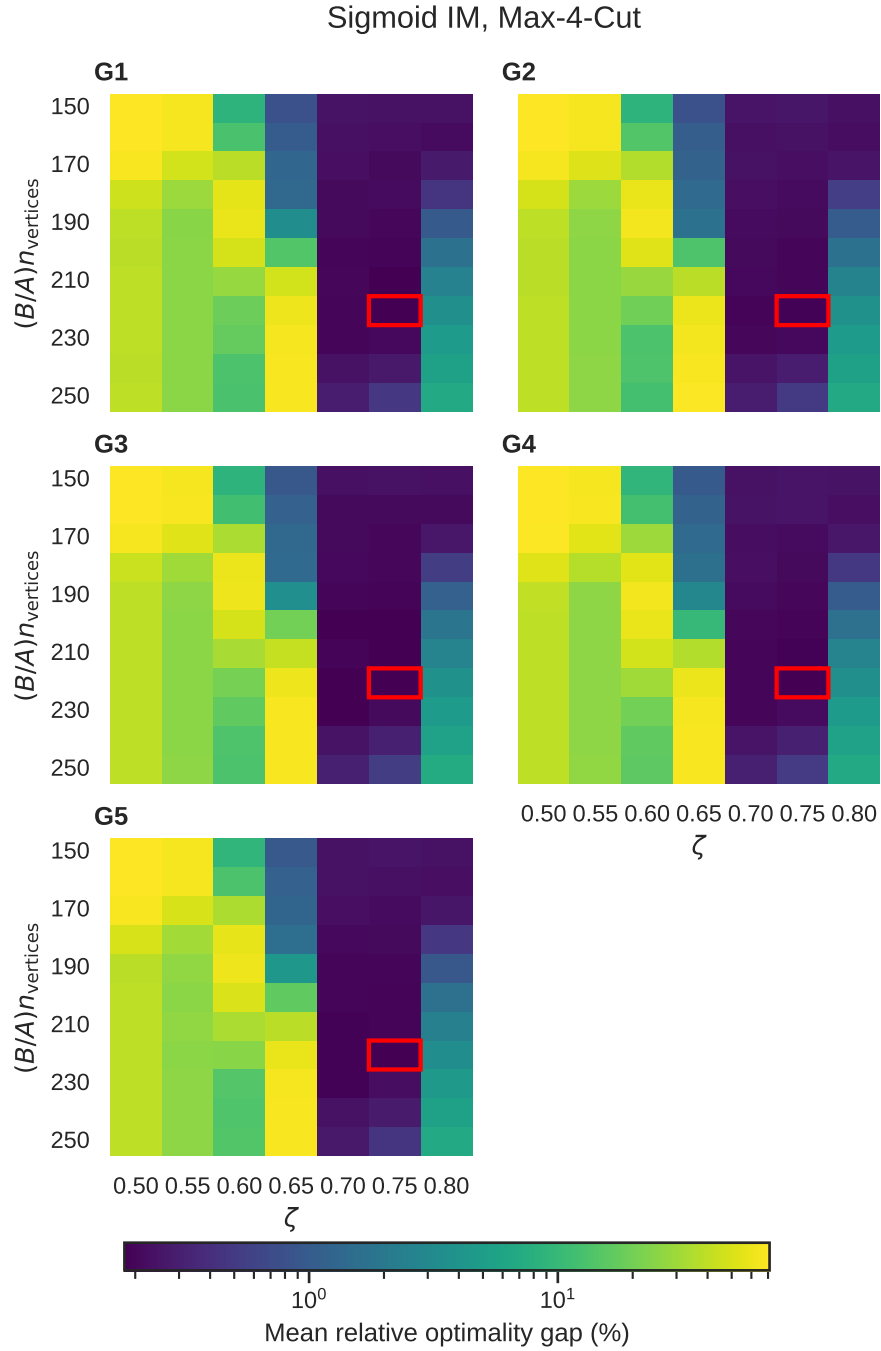


Figure S8.6: Heatmap showing the mean relative optimality gap (%) for the sigmoid IM model over the G1-G5 Max-4-Cut problems as a function of the hyperparameters $(B/A)n_{\text{vertices}}$ and ζ . The values of the mean relative optimality gap are shown in each cell of the heatmap. The hyperparameter configuration with the minimum mean relative optimality gap is marked by a red border.

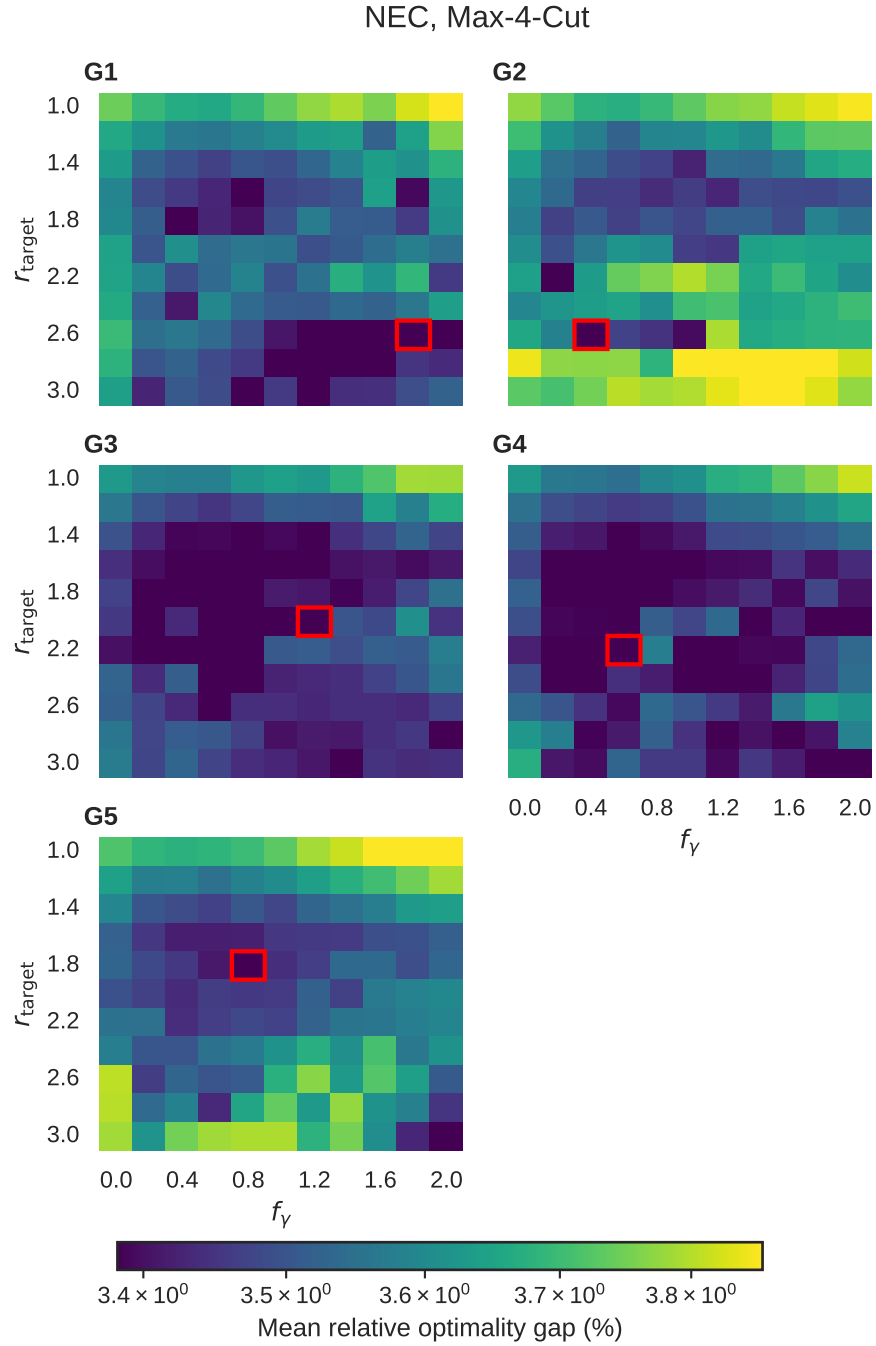


Figure S8.7: Heatmap showing the mean relative optimality gap (%) for the NEC model over the G1-G5 Max-4-Cut problems as a function of the hyperparameters f_γ and r_{target} . The values of the mean relative optimality gap are shown in each cell of the heatmap. The hyperparameter configuration with the minimum mean relative optimality gap is marked by a red border.

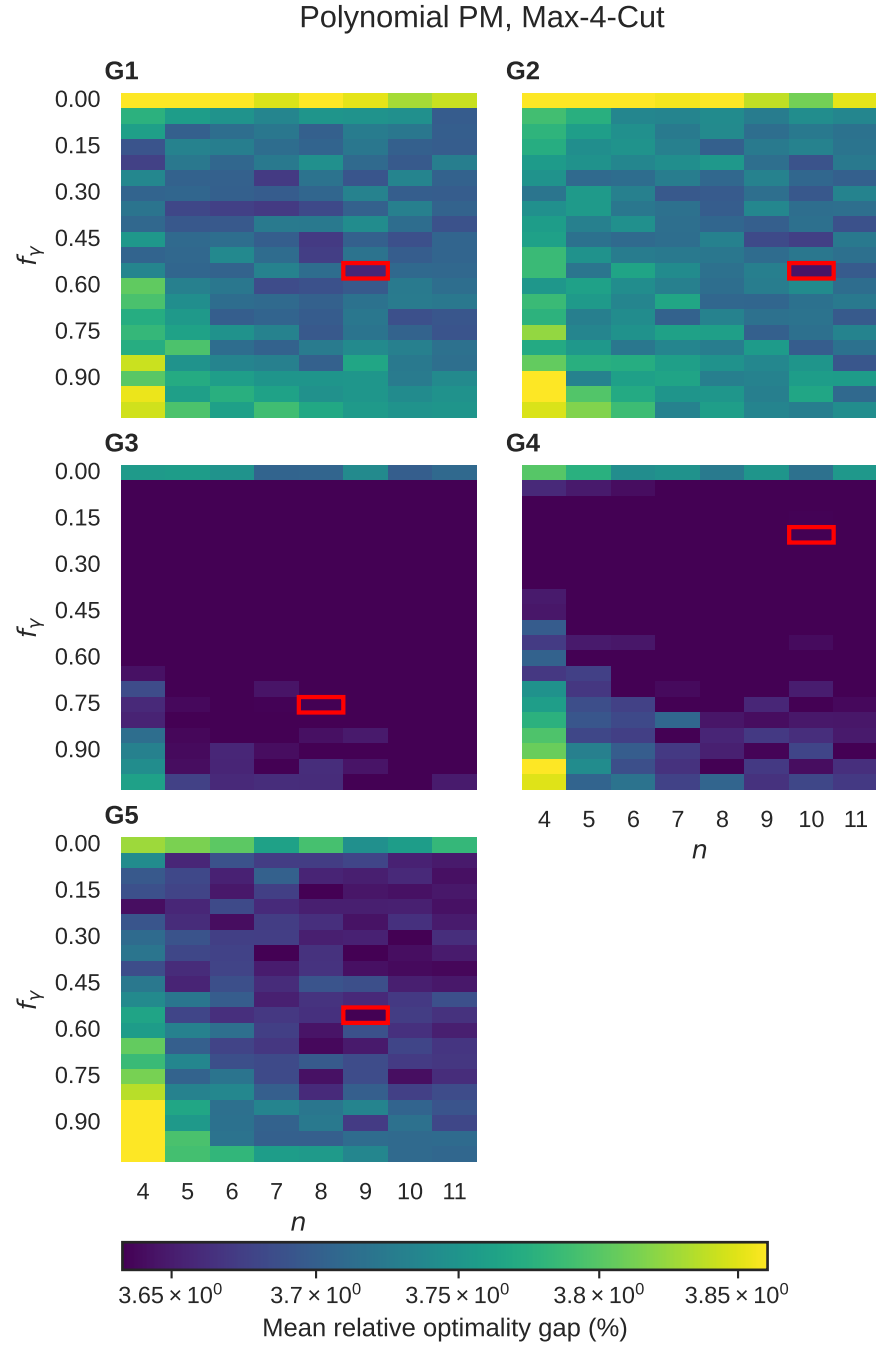


Figure S8.8: Heatmap showing the mean relative optimality gap (%) for the polynomial PM model over the G1-G5 Max-4-Cut problems as a function of the hyperparameters n and f_γ . The values of the mean relative optimality gap are shown in each cell of the heatmap. The hyperparameter configuration with the minimum mean relative optimality gap is marked by a red border.

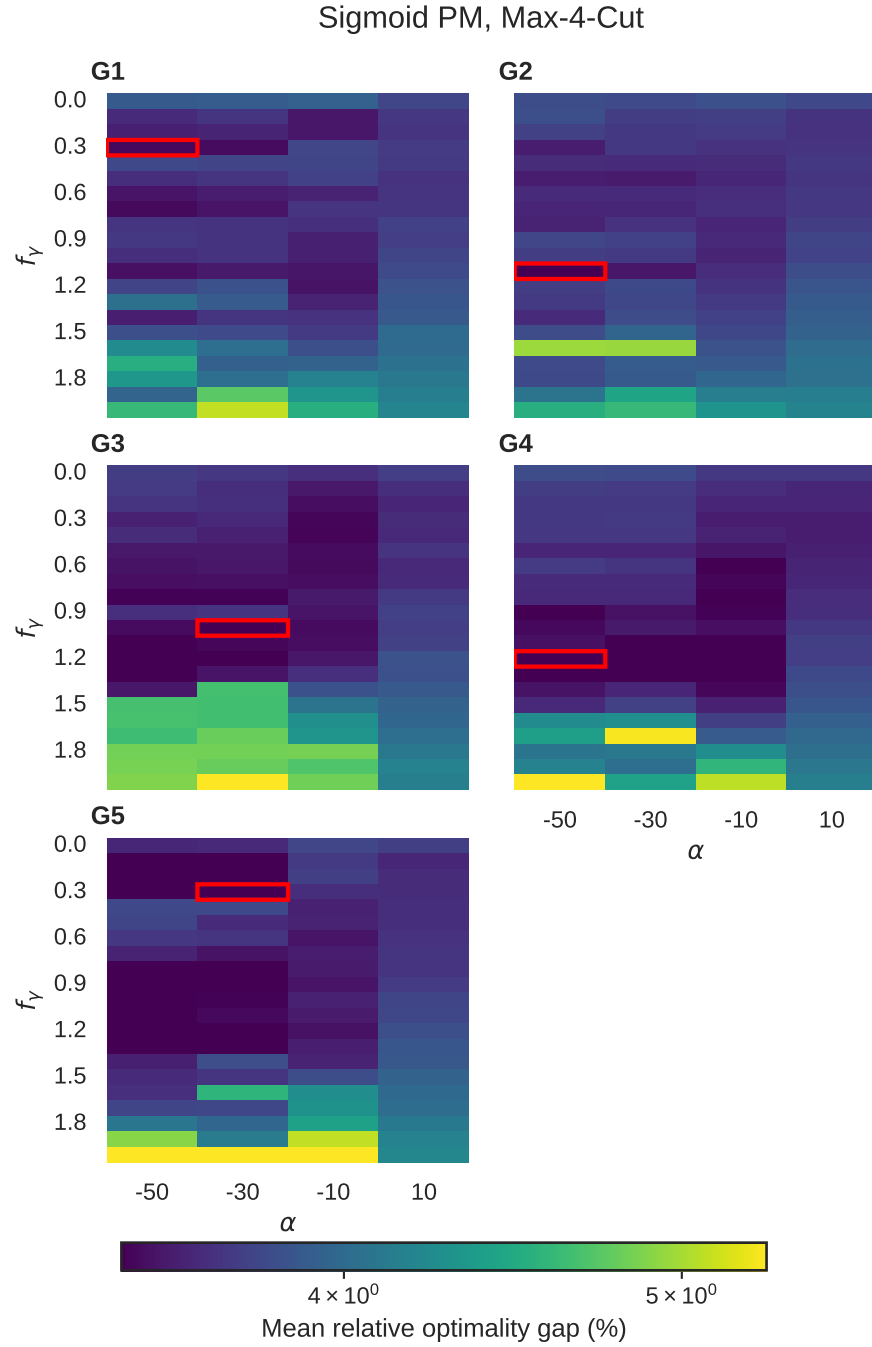


Figure S8.9: Heatmap showing the mean relative optimality gap (%) for the sigmoid PM model over the G1-G5 Max-4-Cut problems as a function of the hyperparameters f_γ and α . The values of the mean relative optimality gap are shown in each cell of the heatmap. The hyperparameter configuration with the minimum mean relative optimality gap is marked by a red border. The sweep of α is limited to a minimum value of -50 to limit computational time, as lower values slows spin dynamics as mentioned in the main text section 2.5.

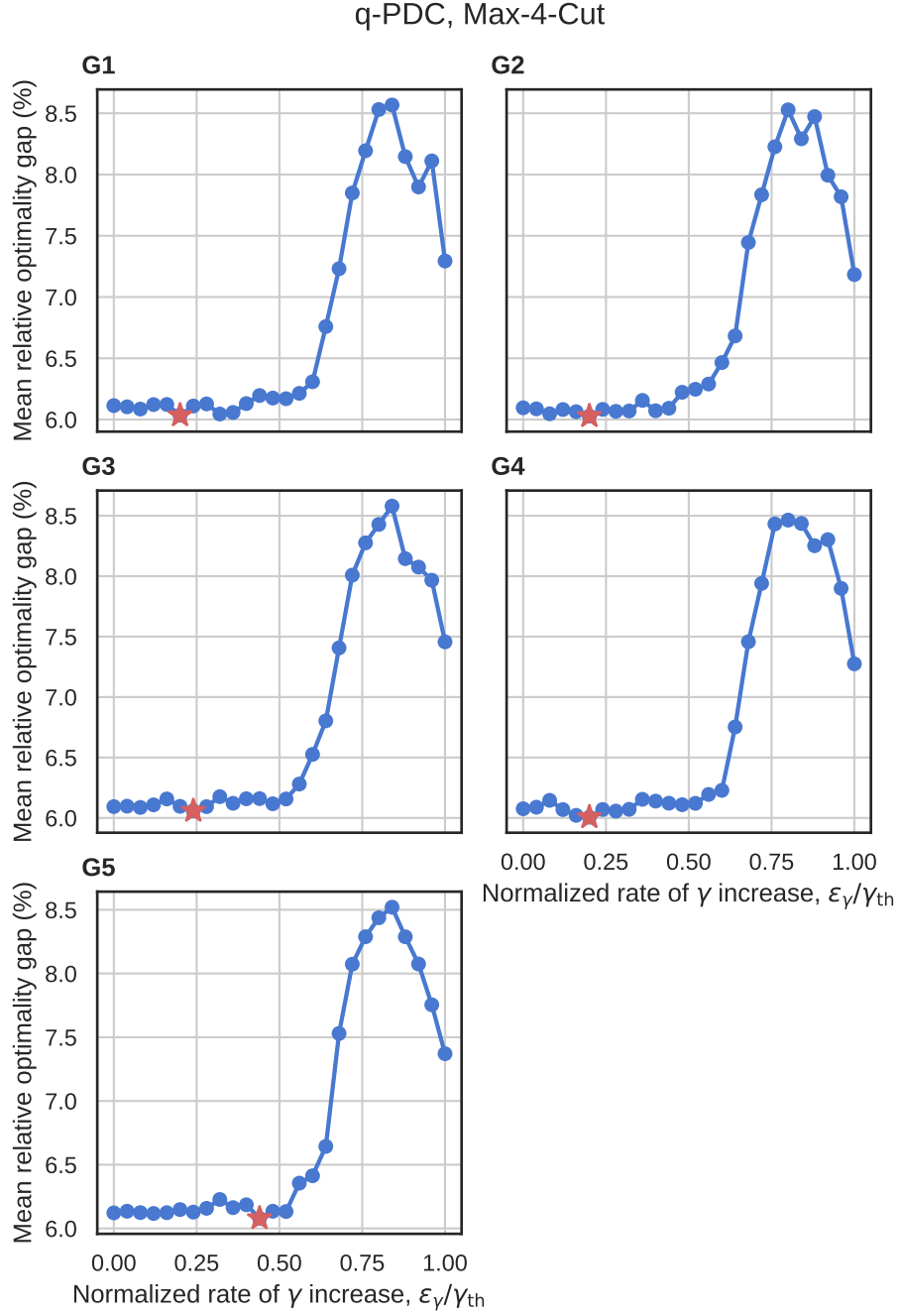


Figure S8.10: Mean relative optimality gap (%) for the q -PDC model over the G1-G5 Max-4-Cut problems as a function of the hyperparameter f_γ . The minimum mean relative optimality gap is marked by a red star.